

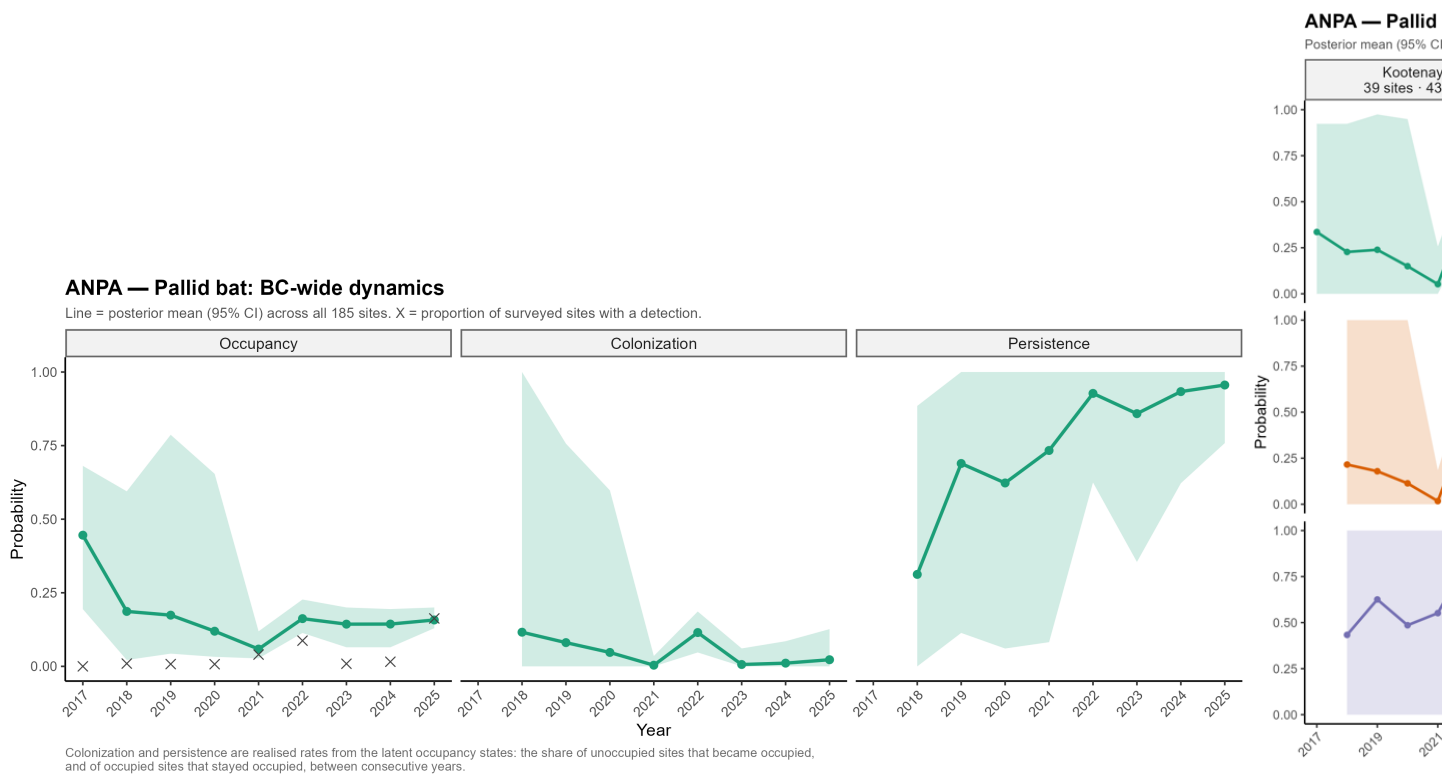
Appendix A

Species and region specific dynamic occupancy results

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0.1 Pallid Bat (*Antrozous pallidus*)



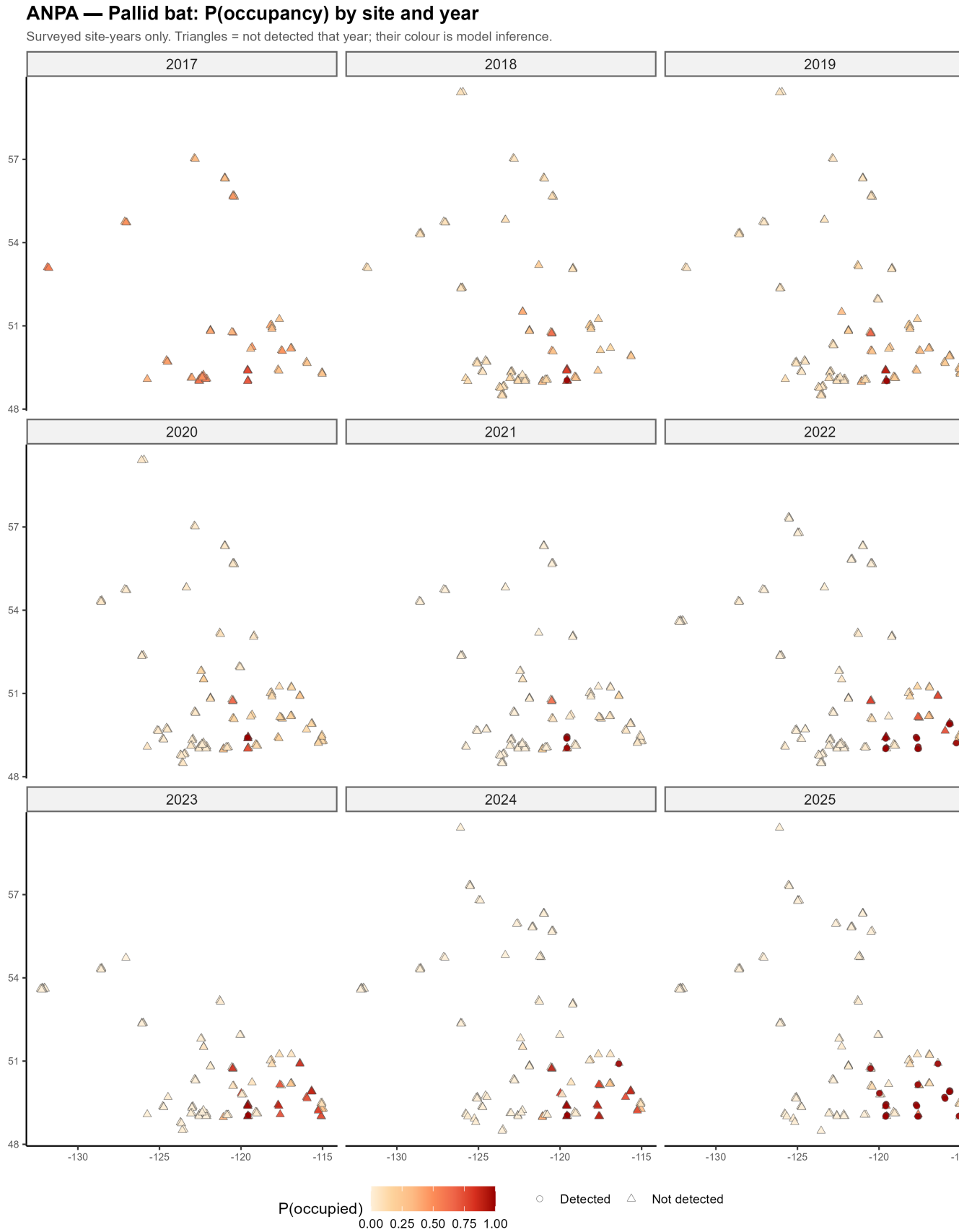
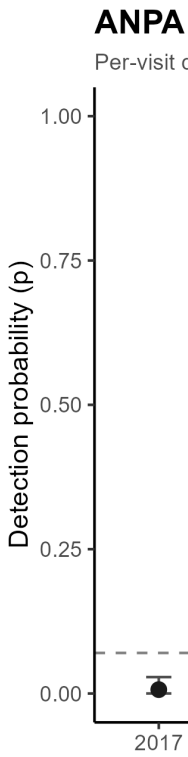
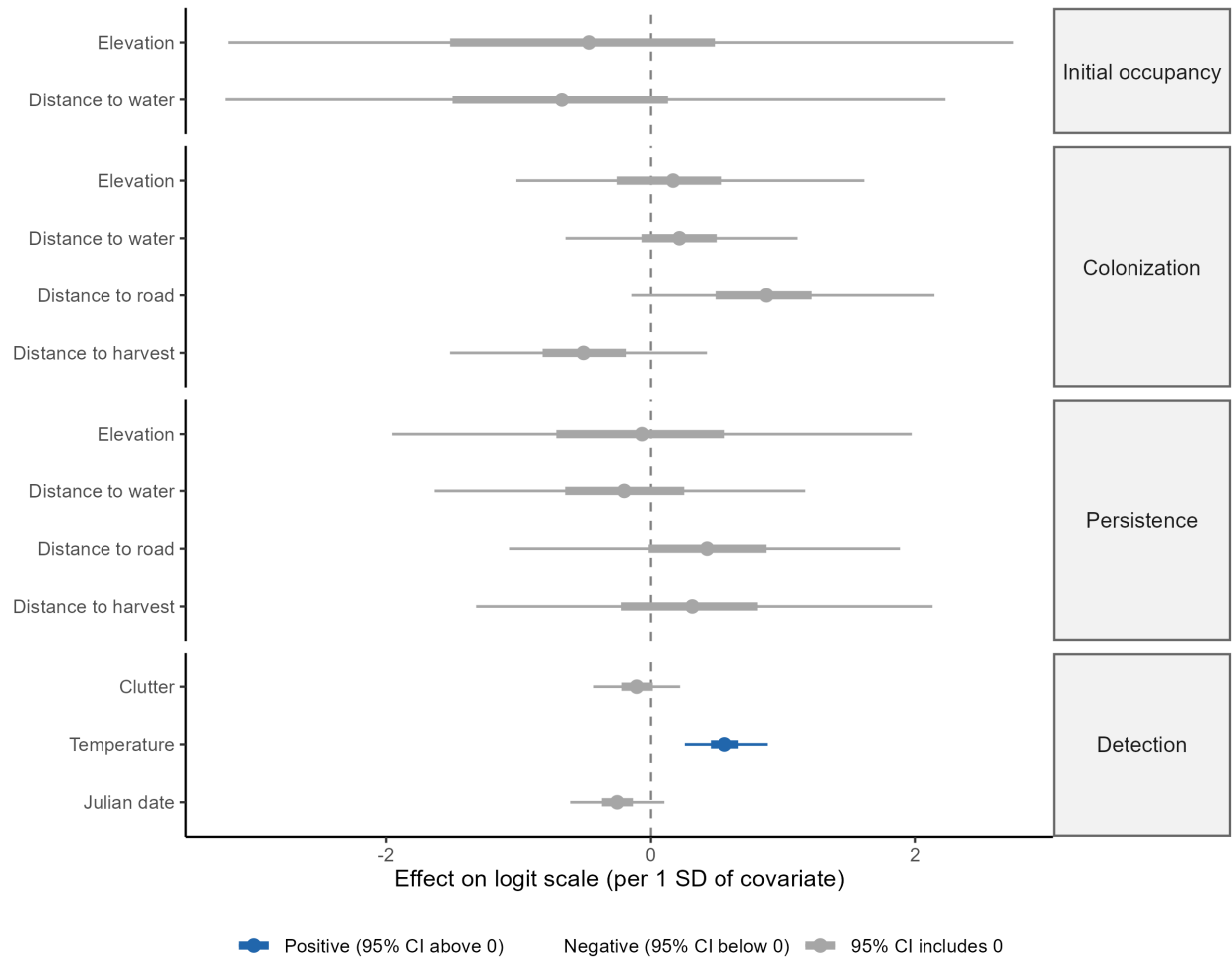


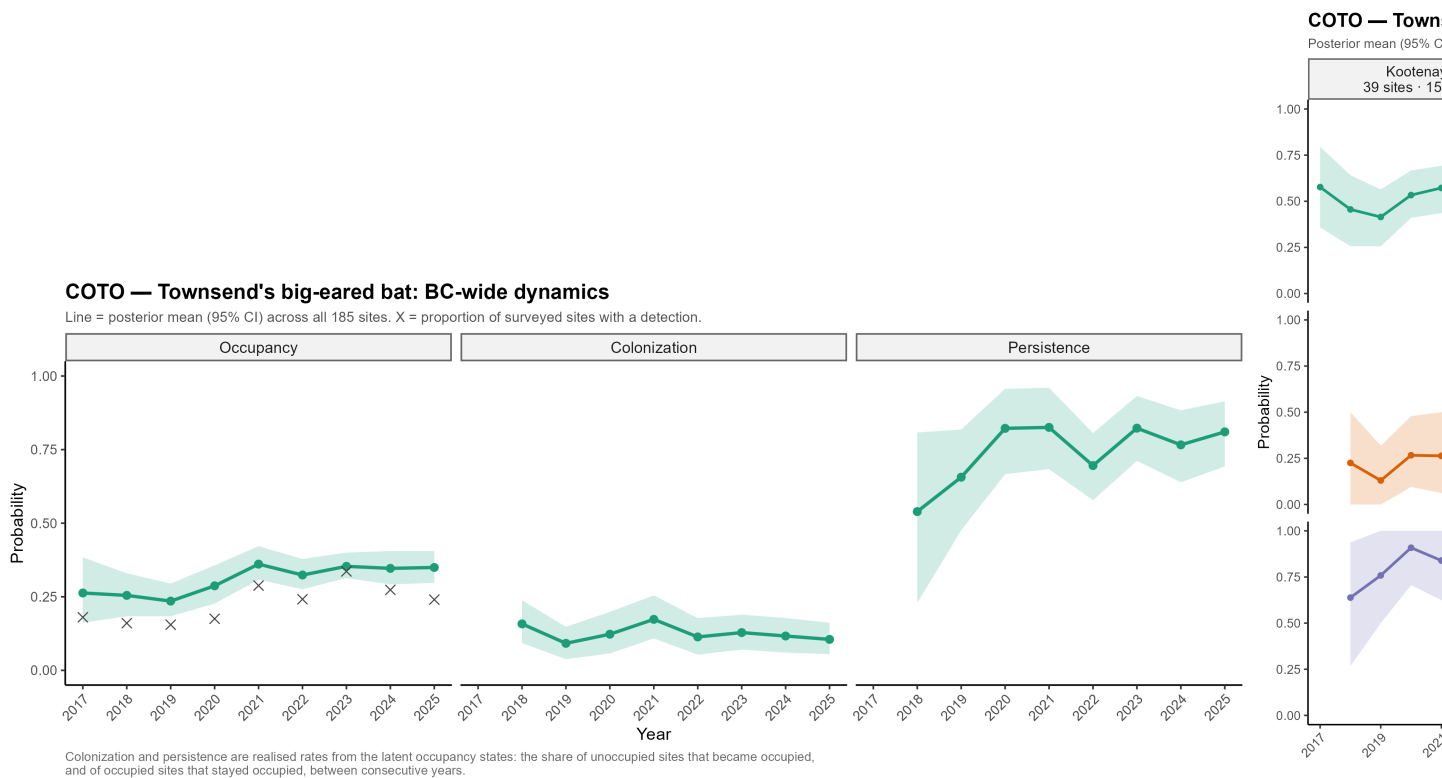
Figure 1: Pallid bat (ANPA): probability of occupancy per site by year. Circles = detected; triangles = never detected (colour is model inference)

ANPA — Pallid bat: covariate effects

Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.



0.2 Townsend’s big-eared bat (*Corynorhinus townsendii*)



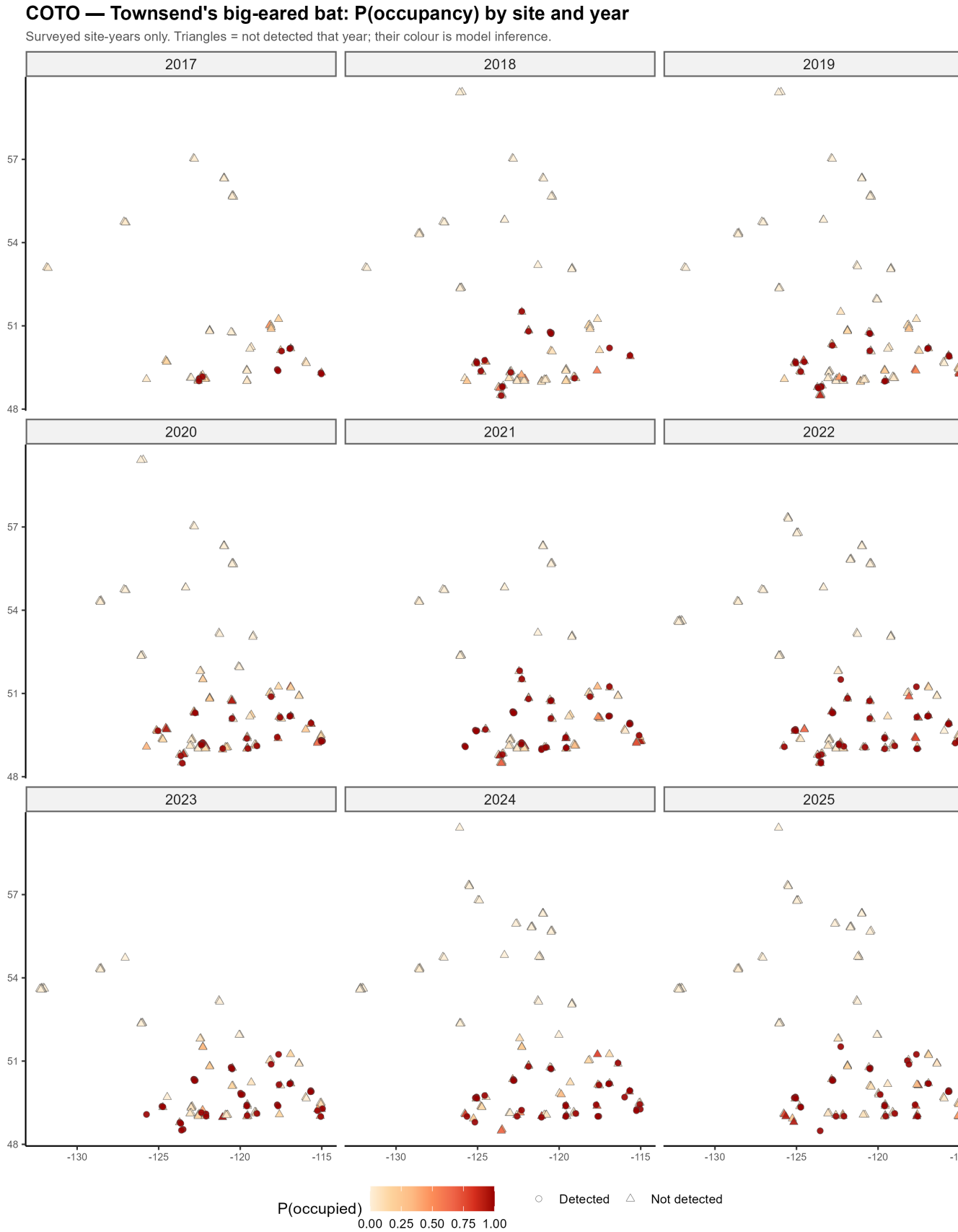
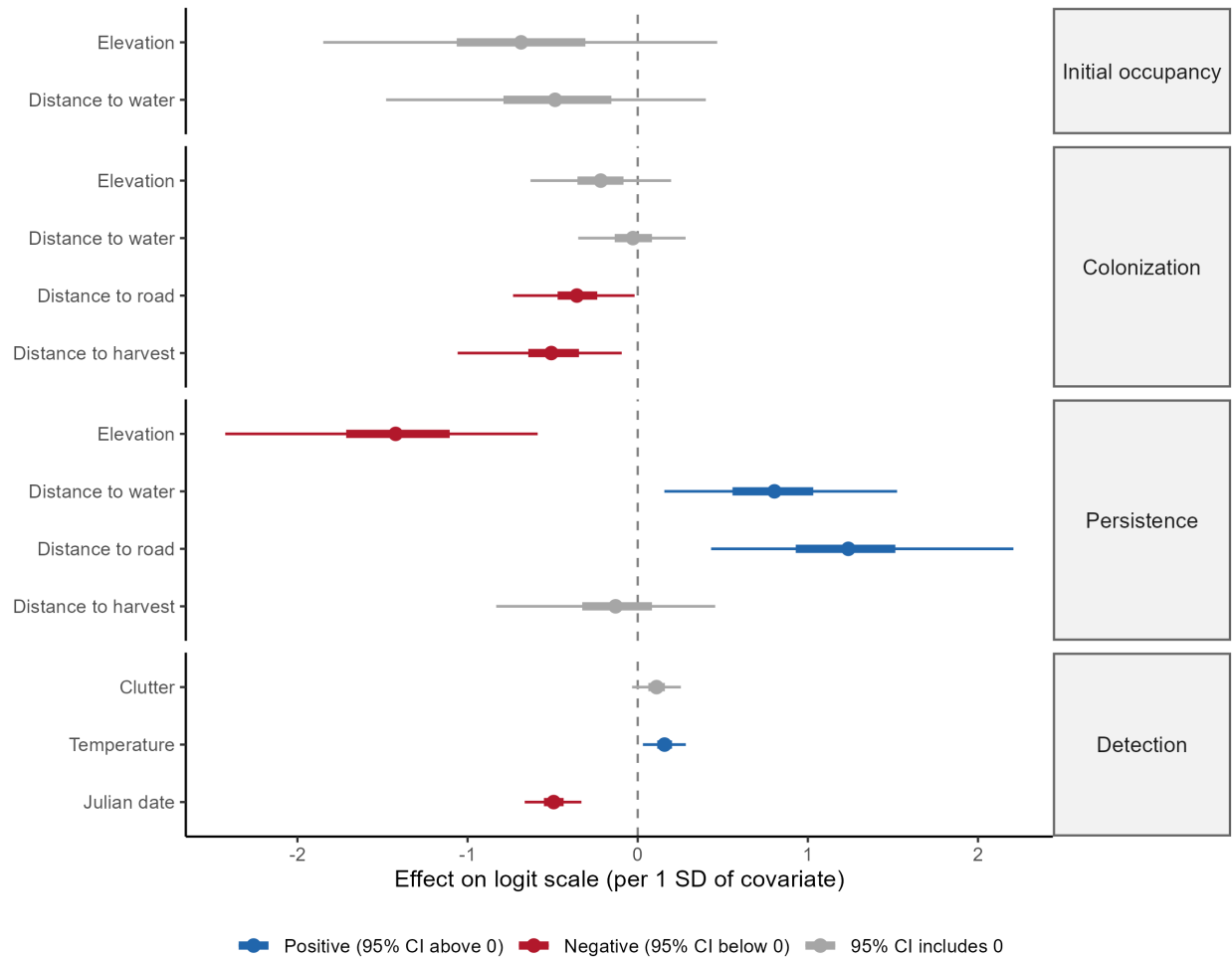


Figure 2: Townsend's big-eared bat (COTO): probability of occupancy per site by year. Circles = detected; triangles = never detected (colour is model inference)

COTO — Townsend's big-eared bat: covariate effects

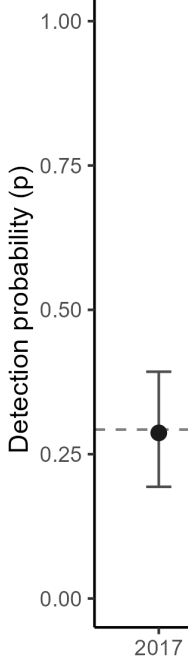
Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.



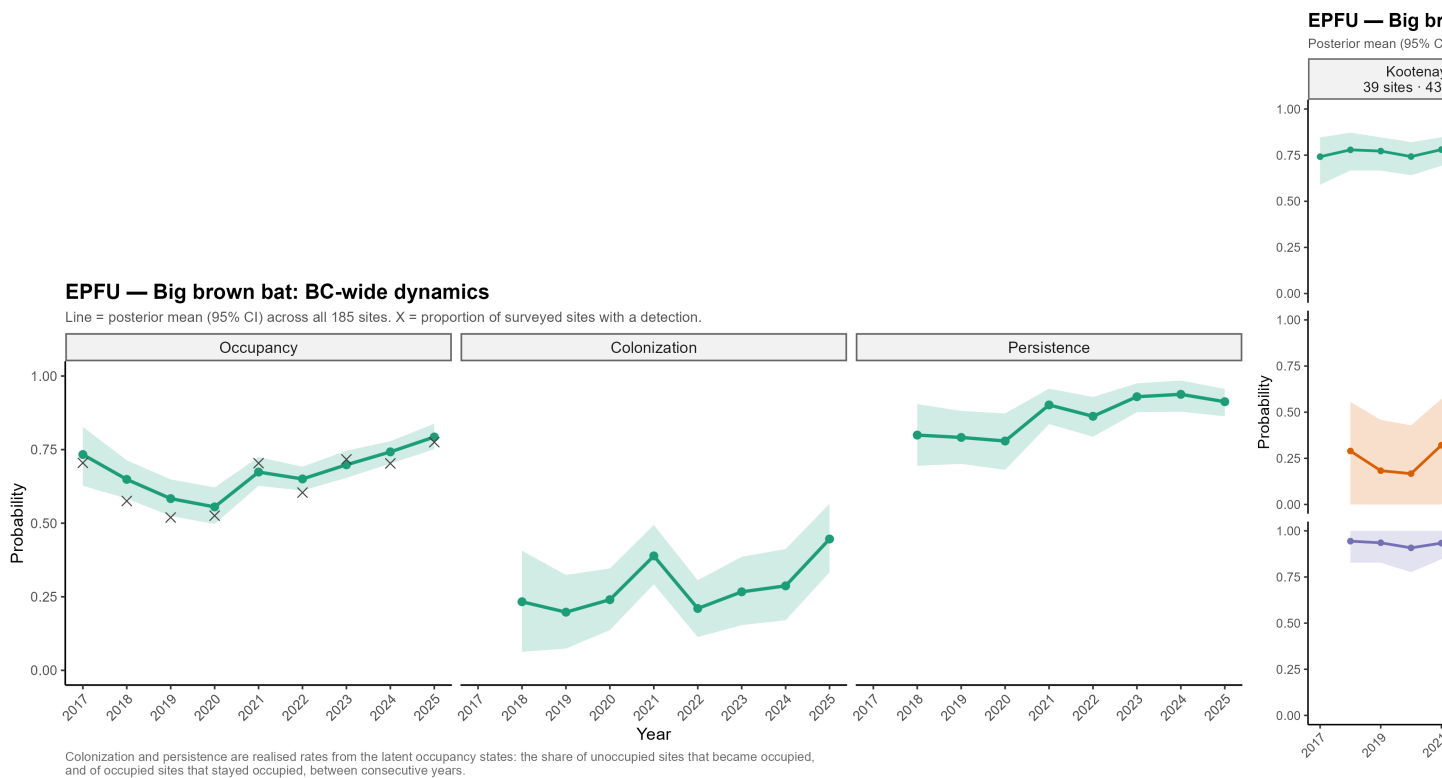
Distances were $\log(x + 1)$ transformed before standardising.

COTO

Per-visit c



0.3 Big brown bat (*Eptesicus fuscus*)



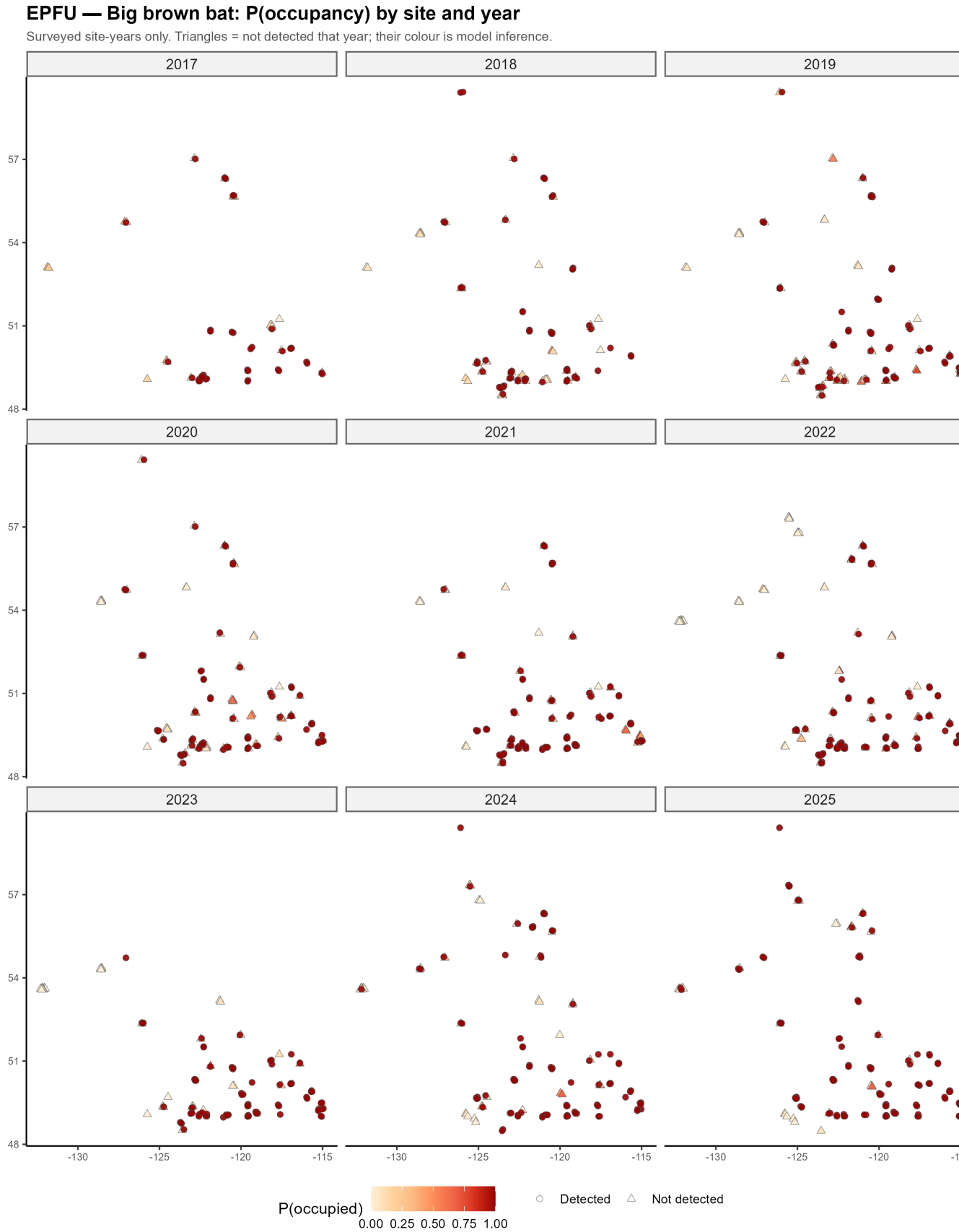
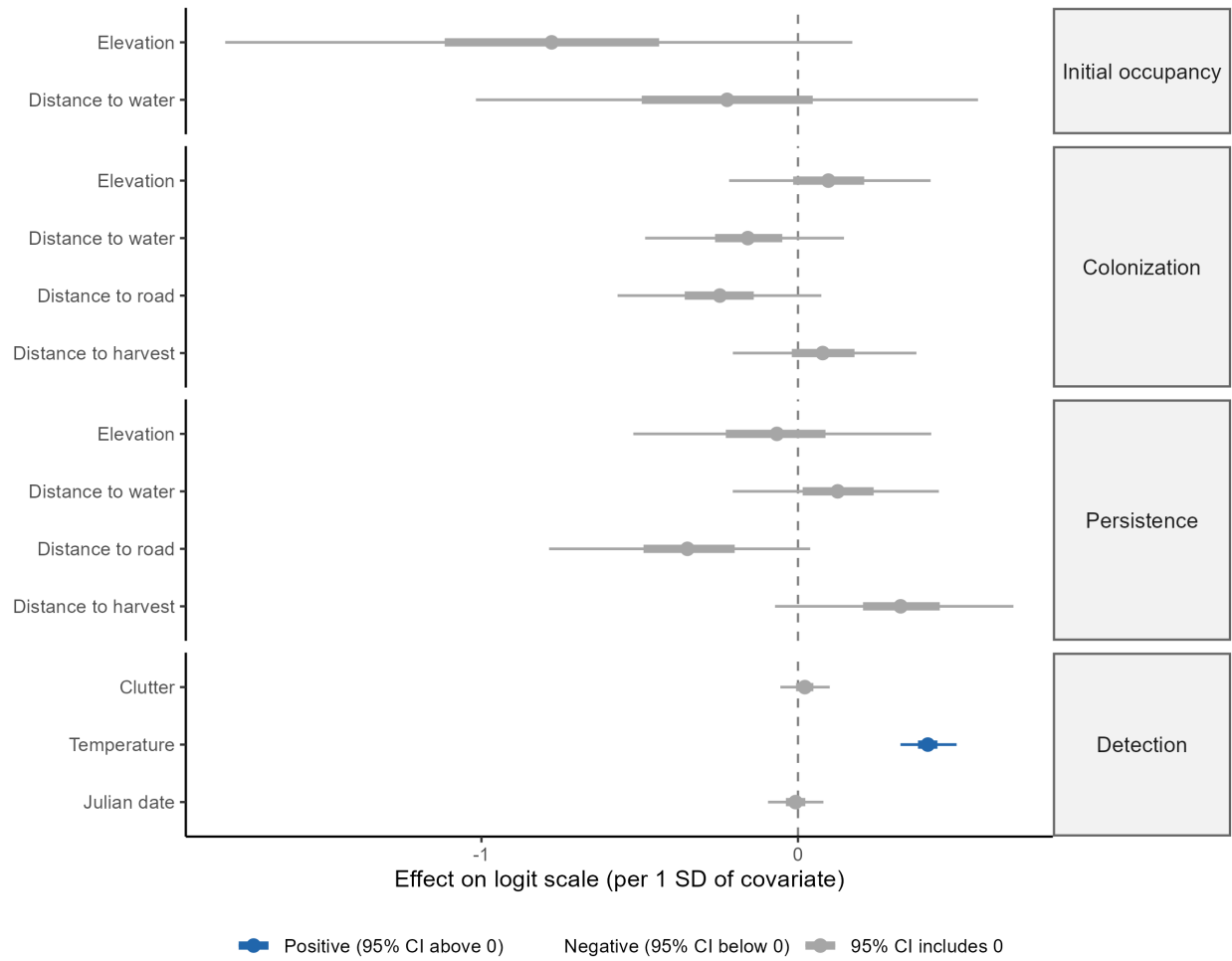


Figure 3: Big brown bat (EPFU): probability of occupancy per site by year. Circles = detected; triangles = never detected (colour is model inference)

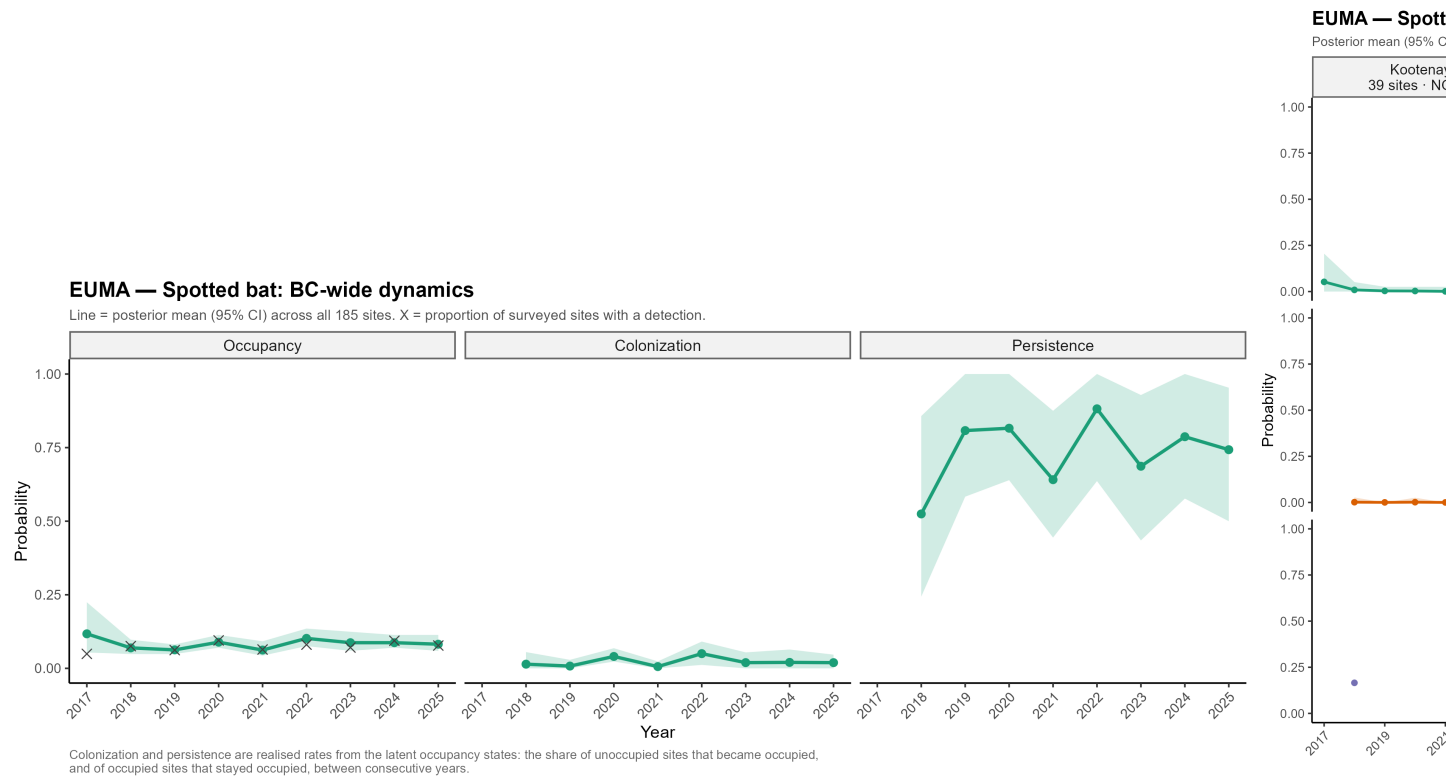
EPFU — Big brown bat: covariate effects

Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.



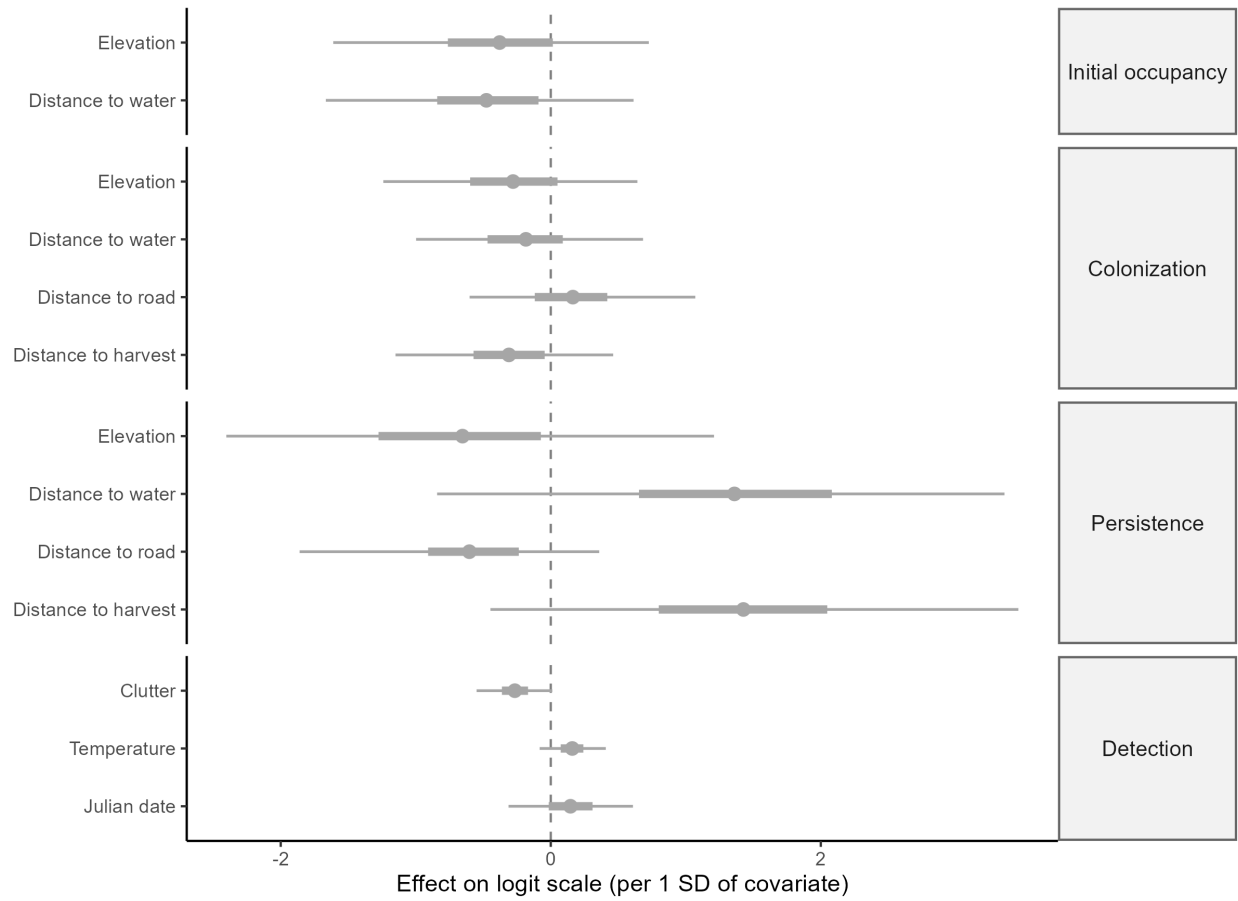
Distances were $\log(x + 1)$ transformed before standardising.

0.4 Spotted bat (*Euderma maculatum*)



EUMA — Spotted bat: covariate effects

Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.



Positive (95% CI above 0)

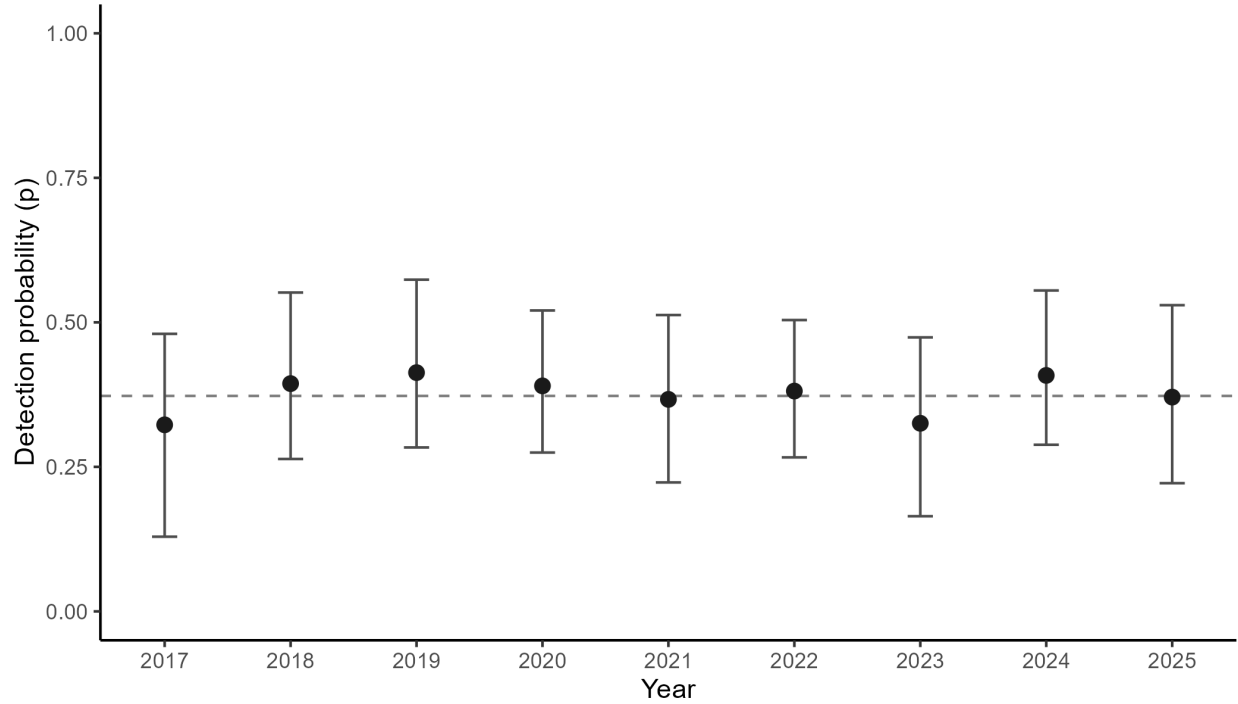
Negative (95% CI below 0)

95% CI includes 0

Distances were $\log(x + 1)$ transformed before standardising.

EUMA — Spotted bat: detection probability by year

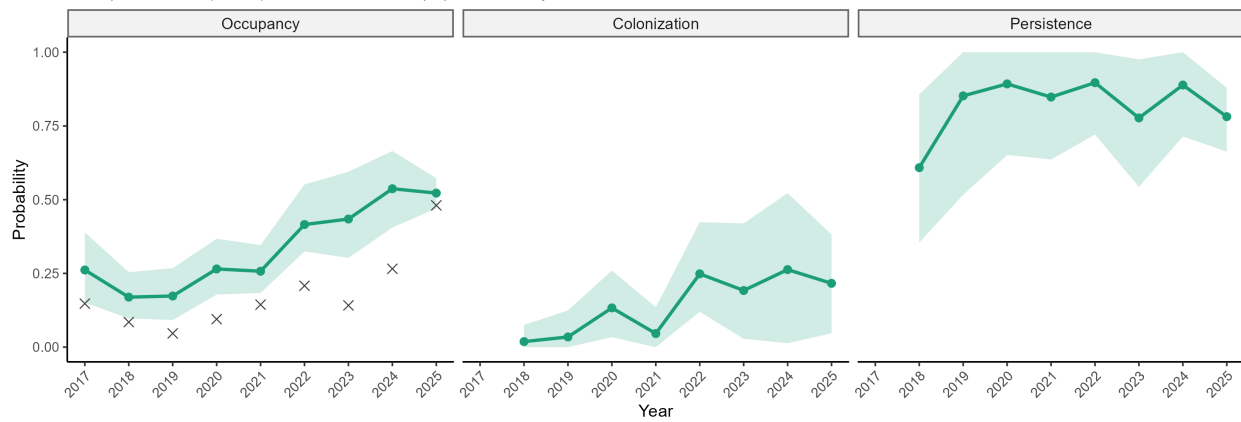
Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.



0.5 Easter red bat (*Lasiurus borealis*)

LABO — Eastern red bat: BC-wide dynamics

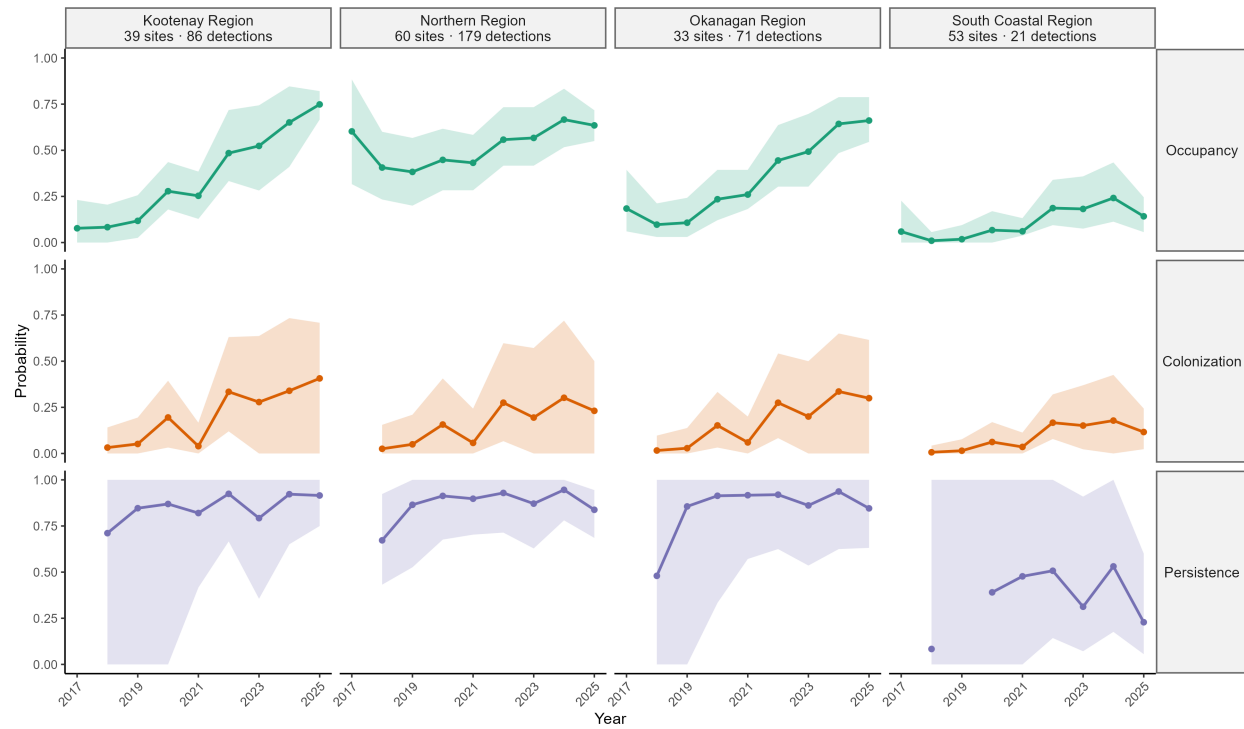
Line = posterior mean (95% CI) across all 185 sites. X = proportion of surveyed sites with a detection.



Colonization and persistence are realised rates from the latent occupancy states: the share of unoccupied sites that became occupied, and of occupied sites that stayed occupied, between consecutive years.

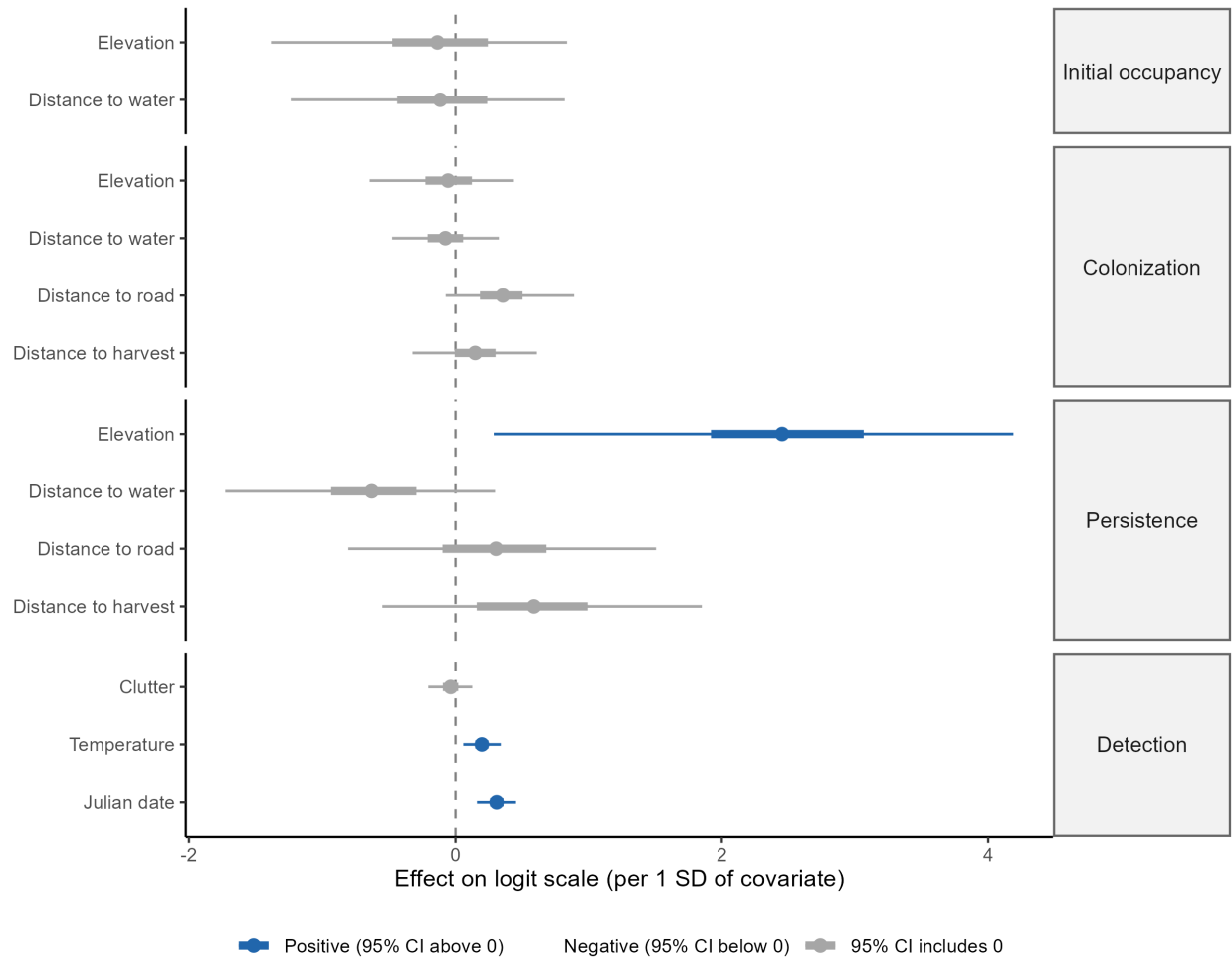
LABO — Eastern red bat: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



LABO — Eastern red bat: covariate effects

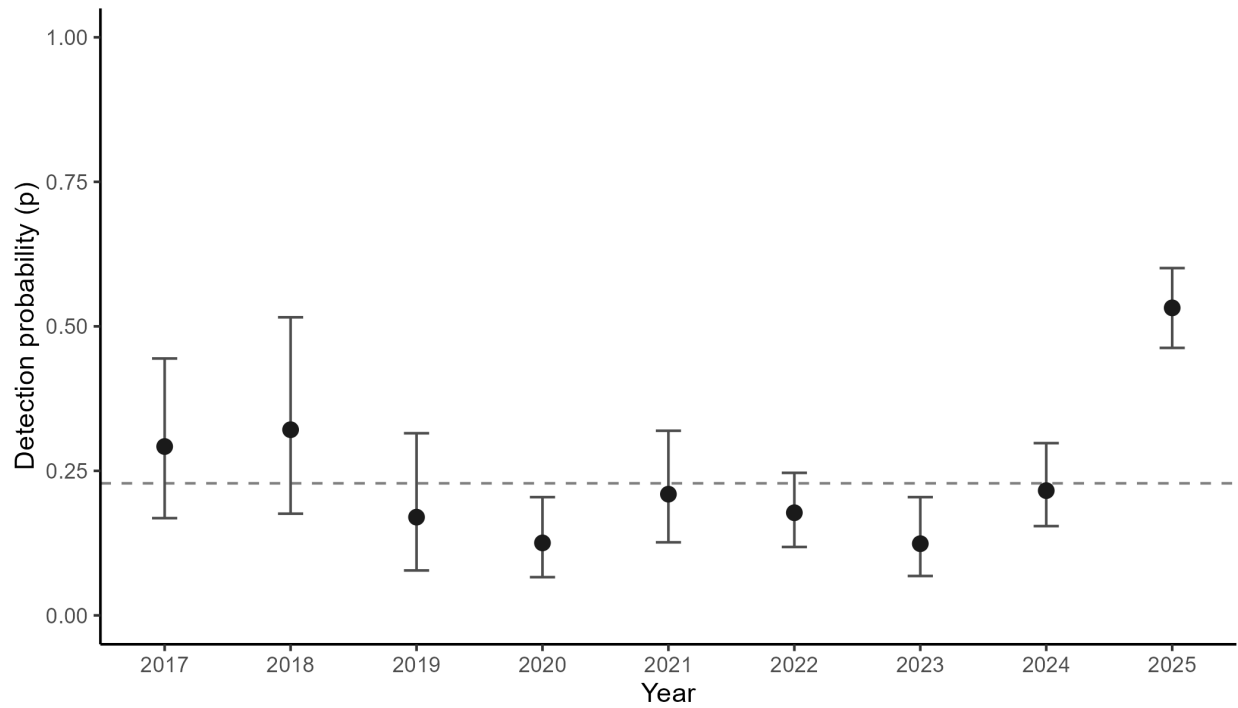
Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.



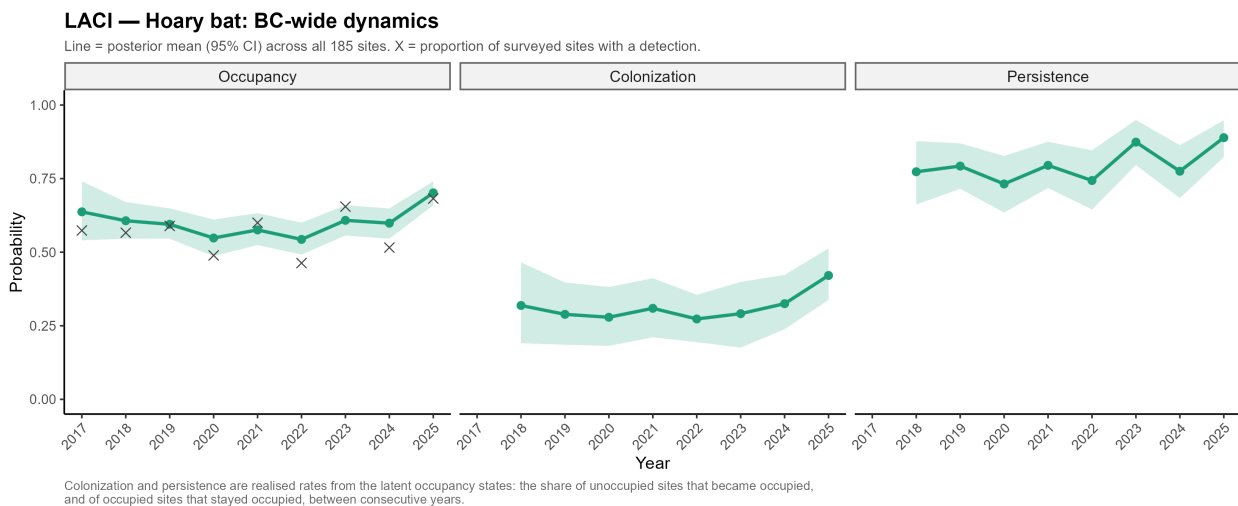
Distances were $\log(x + 1)$ transformed before standardising.

LABO — Eastern red bat: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.

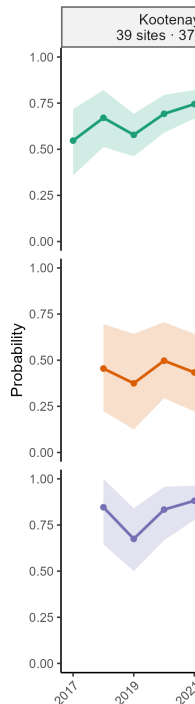


0.6 Hoary Bat (*Lasiurus cinereus*)



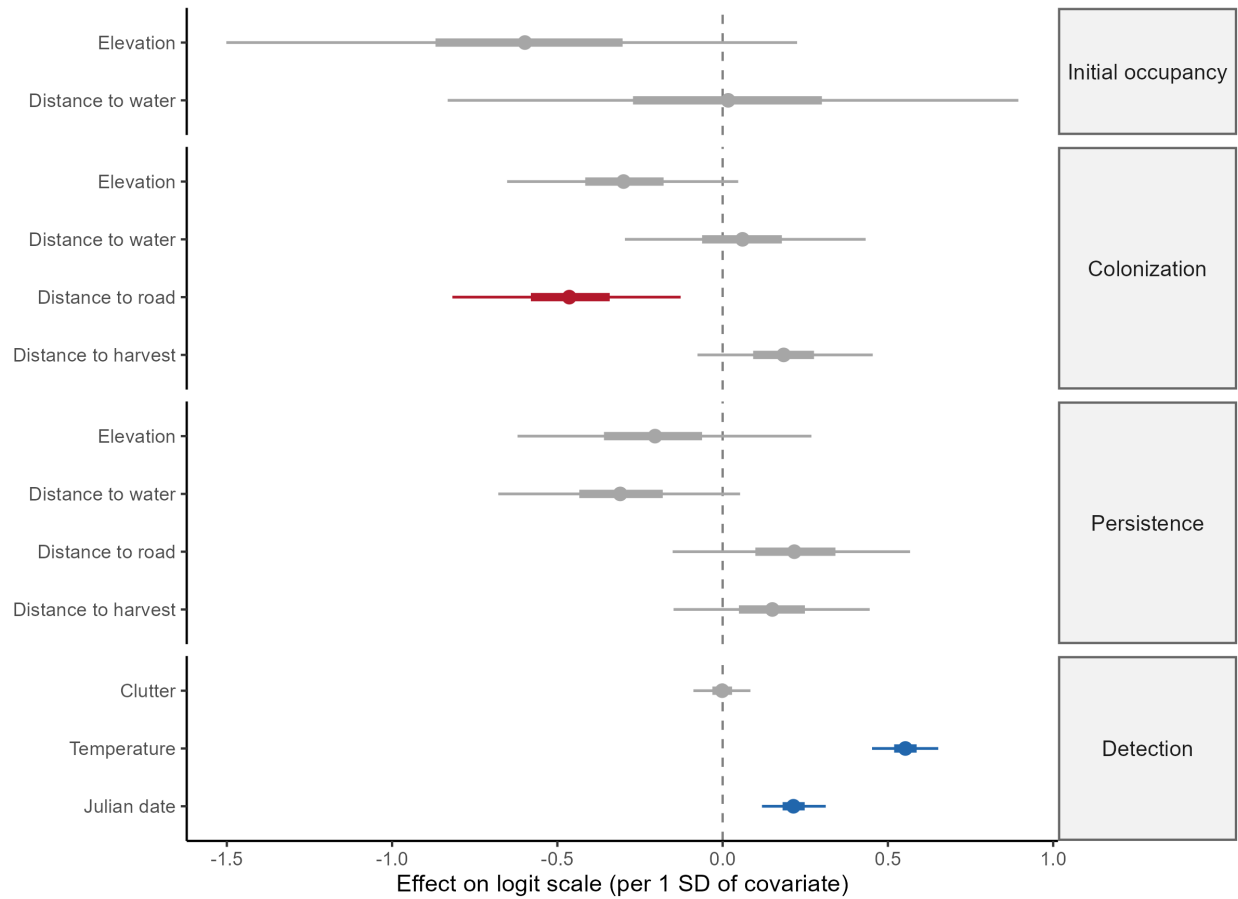
LACI — Hoary

Posterior mean (95% CI)



LACI — Hoary bat: covariate effects

Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.

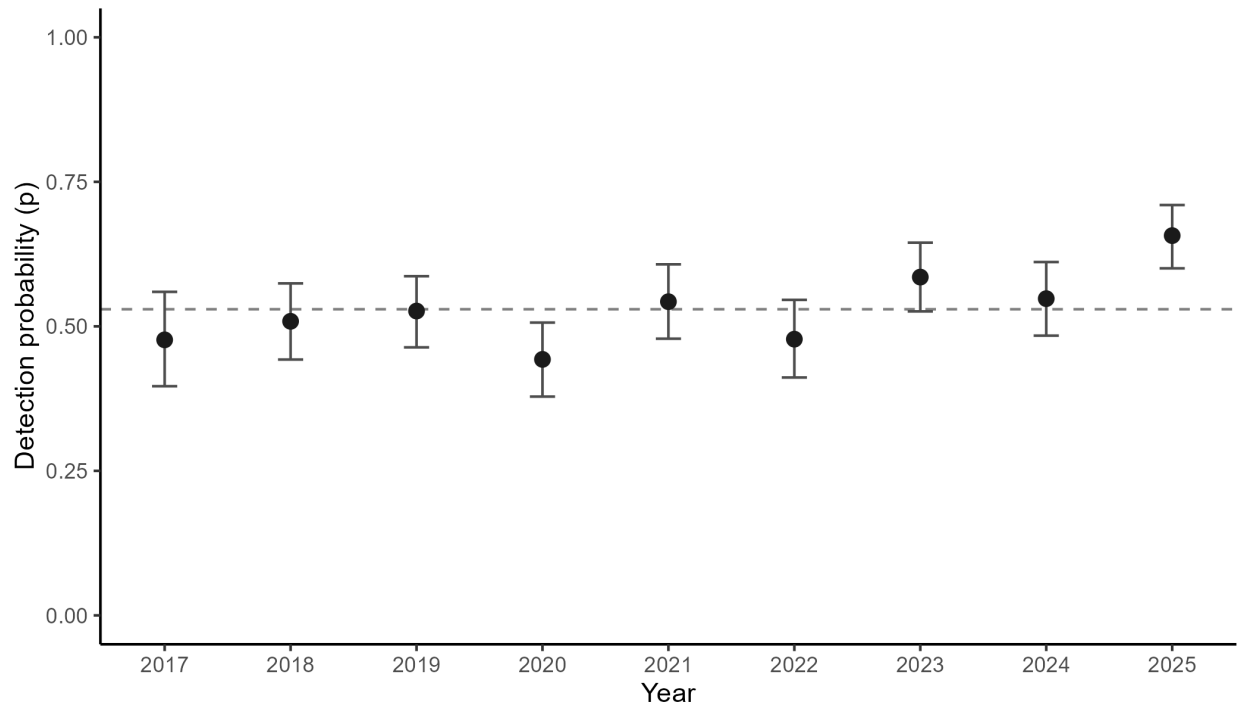


■ Positive (95% CI above 0)
 ■ Negative (95% CI below 0)
 ■ 95% CI includes 0

Distances were $\log(x + 1)$ transformed before standardising.

LACI — Hoary bat: detection probability by year

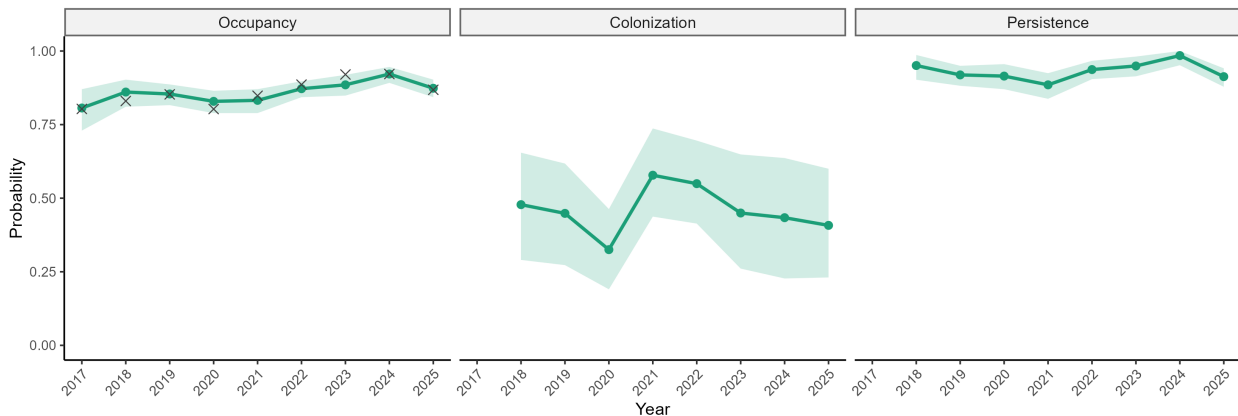
Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.



0.7 Silver-haired bat (*Lasionycteris noctivagans*)

LANO — Silver-haired bat: BC-wide dynamics

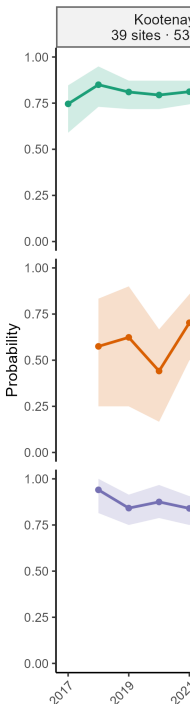
Line = posterior mean (95% CI) across all 185 sites. X = proportion of surveyed sites with a detection.



Colonization and persistence are realised rates from the latent occupancy states: the share of unoccupied sites that became occupied, and of occupied sites that stayed occupied, between consecutive years.

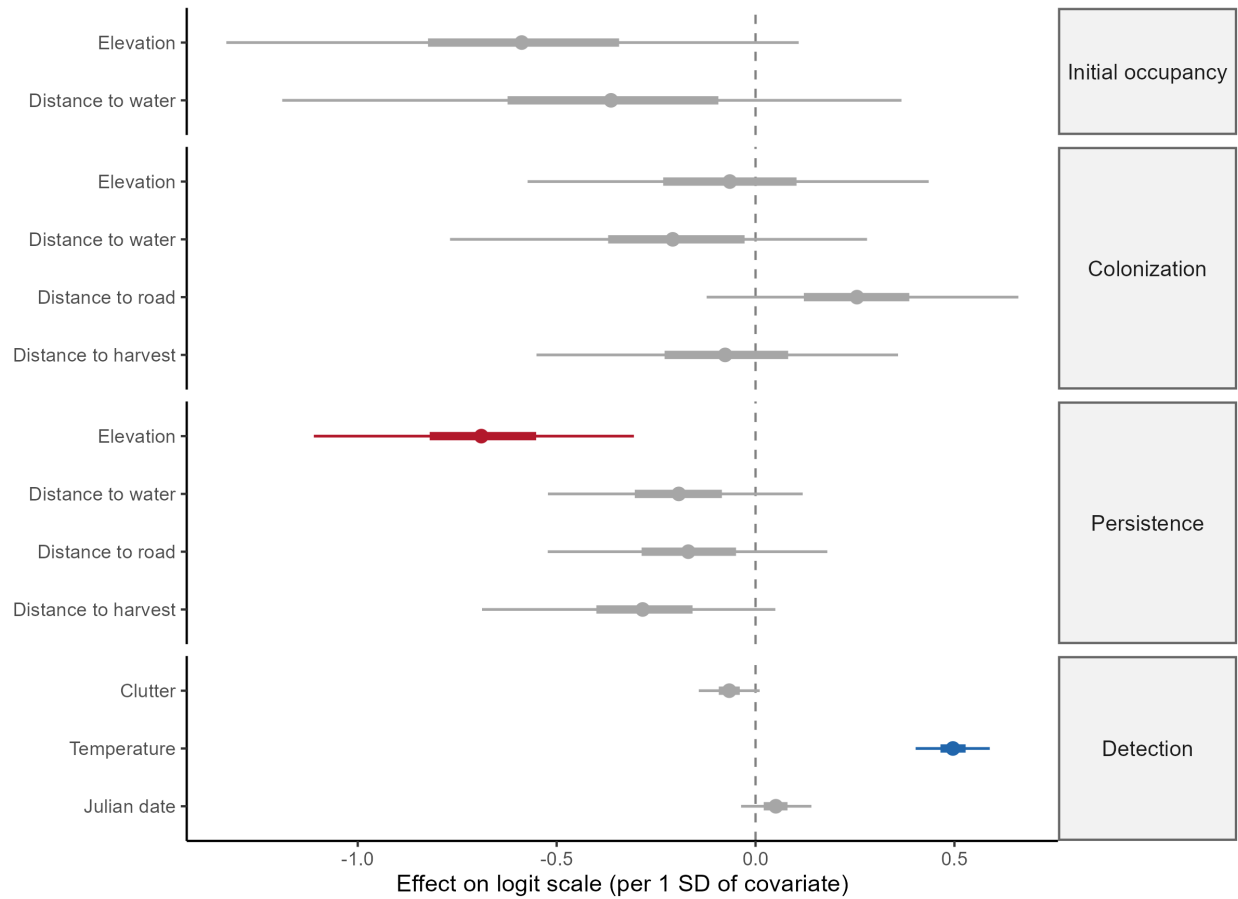
LANO — Silver

Posterior mean (95% CI)



LANO — Silver-haired bat: covariate effects

Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.

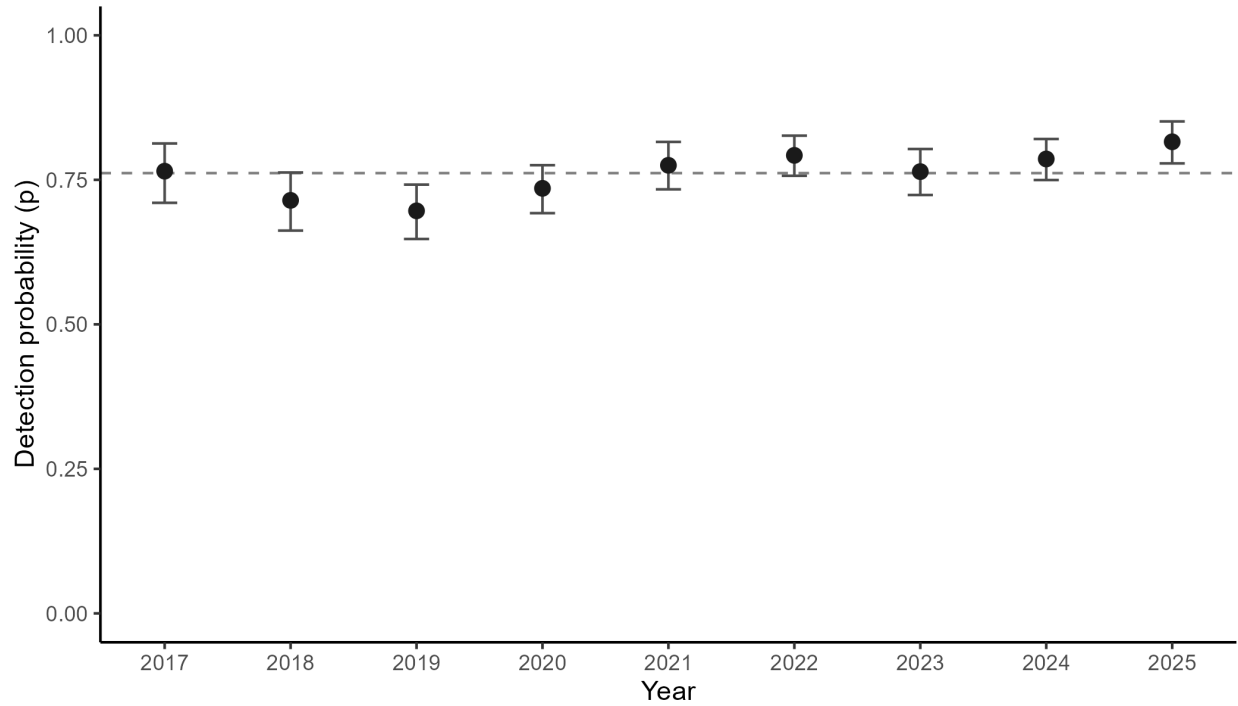


■ Positive (95% CI above 0)
 ■ Negative (95% CI below 0)
 ■ 95% CI includes 0

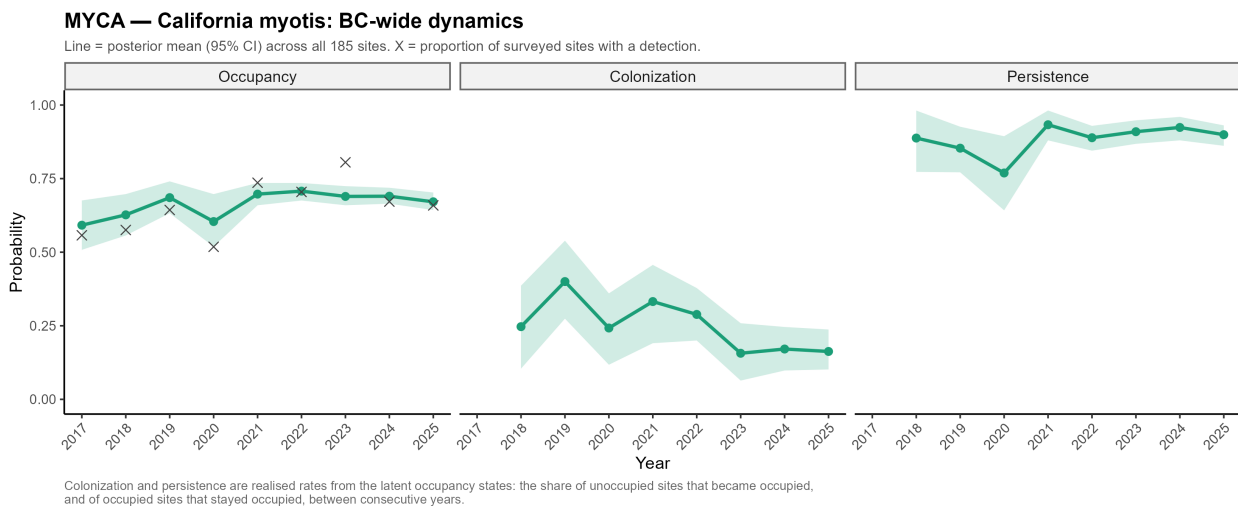
Distances were $\log(x + 1)$ transformed before standardising.

LANO — Silver-haired bat: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.

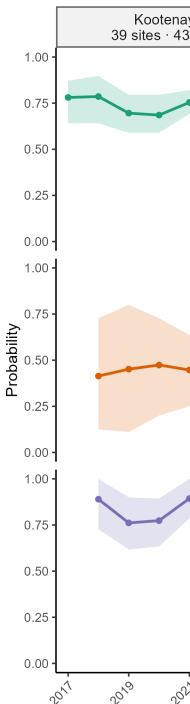


0.8 California Myotis (*Myotis californicus*)



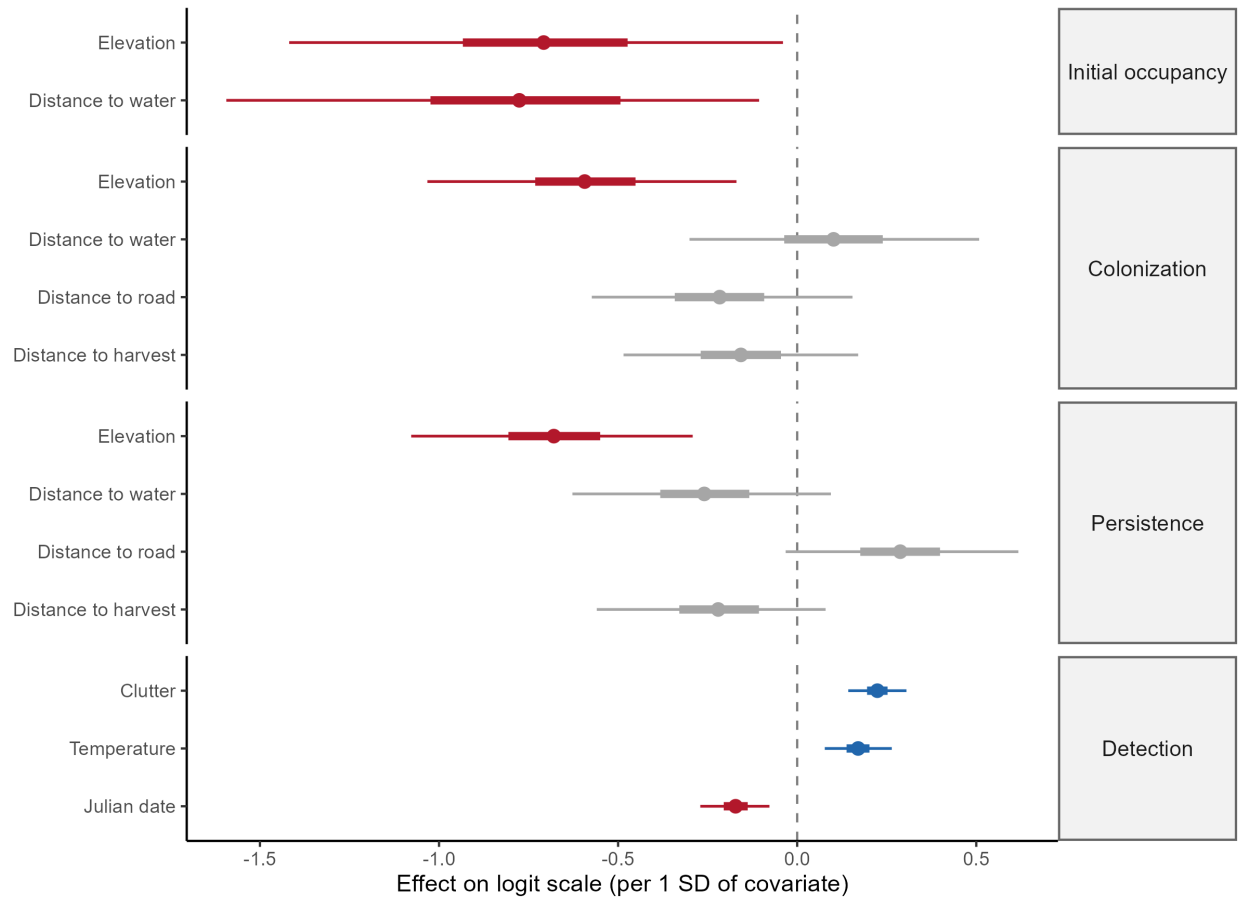
MYCA — California myotis: Kootenai

Posterior mean (95% CI) across all 39 sites.



MYCA — California myotis: covariate effects

Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.

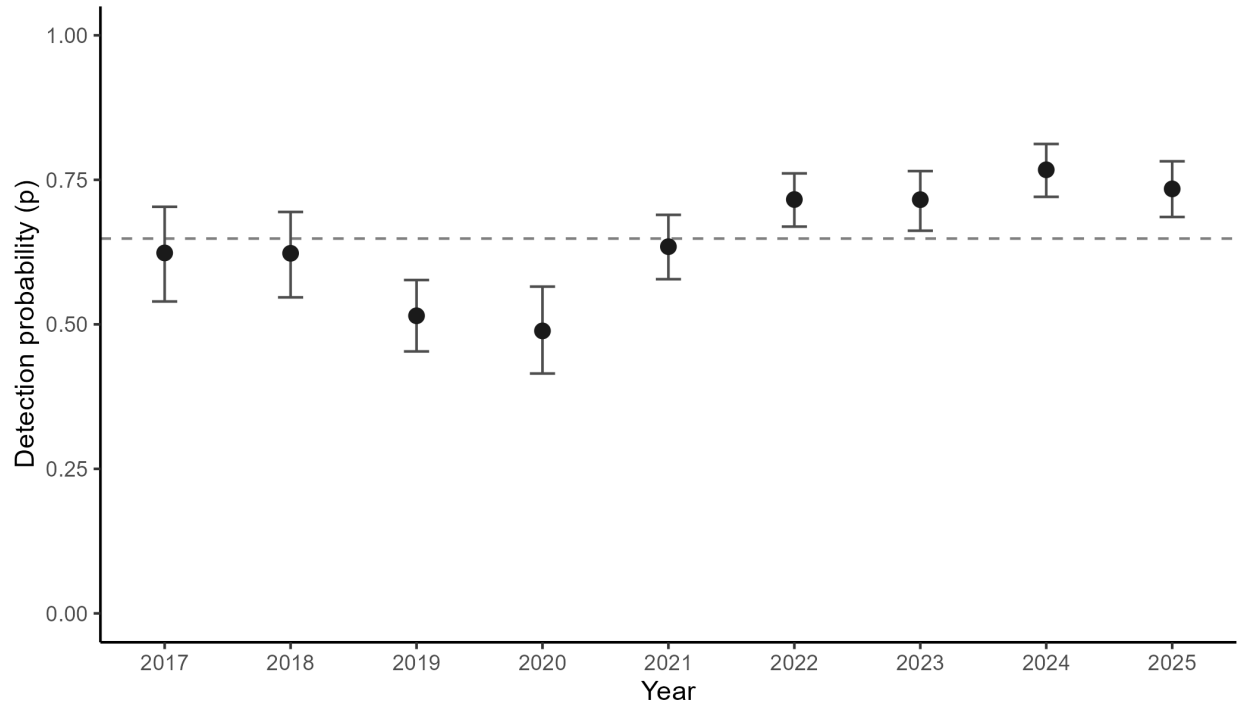


■ Positive (95% CI above 0)
 ■ Negative (95% CI below 0)
 ■ 95% CI includes 0

Distances were $\log(x + 1)$ transformed before standardising.

MYCA — California myotis: detection probability by year

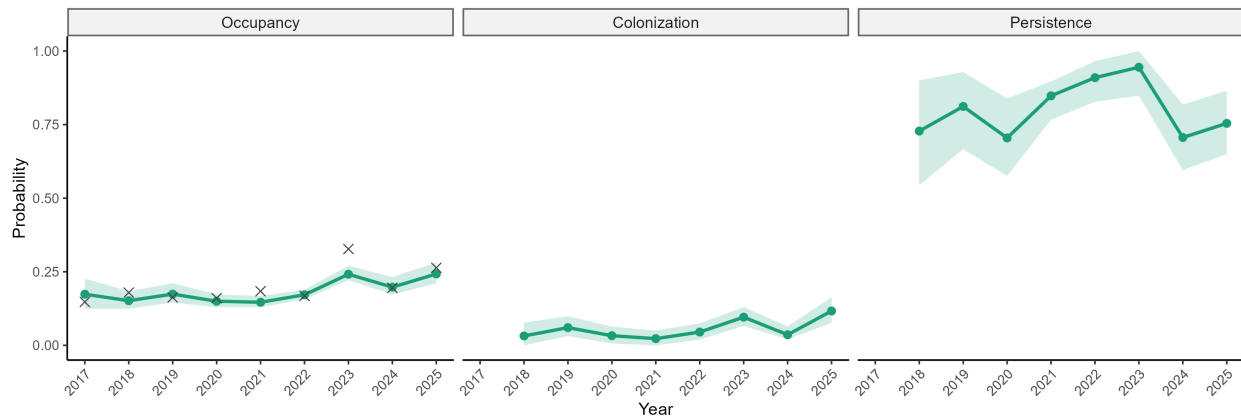
Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.



0.9 Western small-footed Myotis (*Myotis ciliolabrum*)

MYCI — Western small-footed myotis: BC-wide dynamics

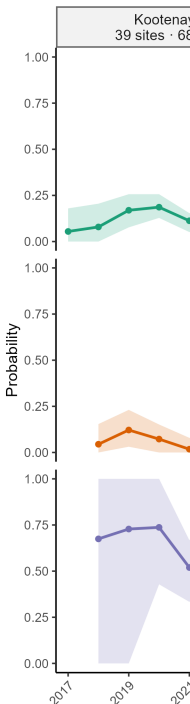
Line = posterior mean (95% CI) across all 185 sites. X = proportion of surveyed sites with a detection.



Colonization and persistence are realised rates from the latent occupancy states: the share of unoccupied sites that became occupied, and of occupied sites that stayed occupied, between consecutive years.

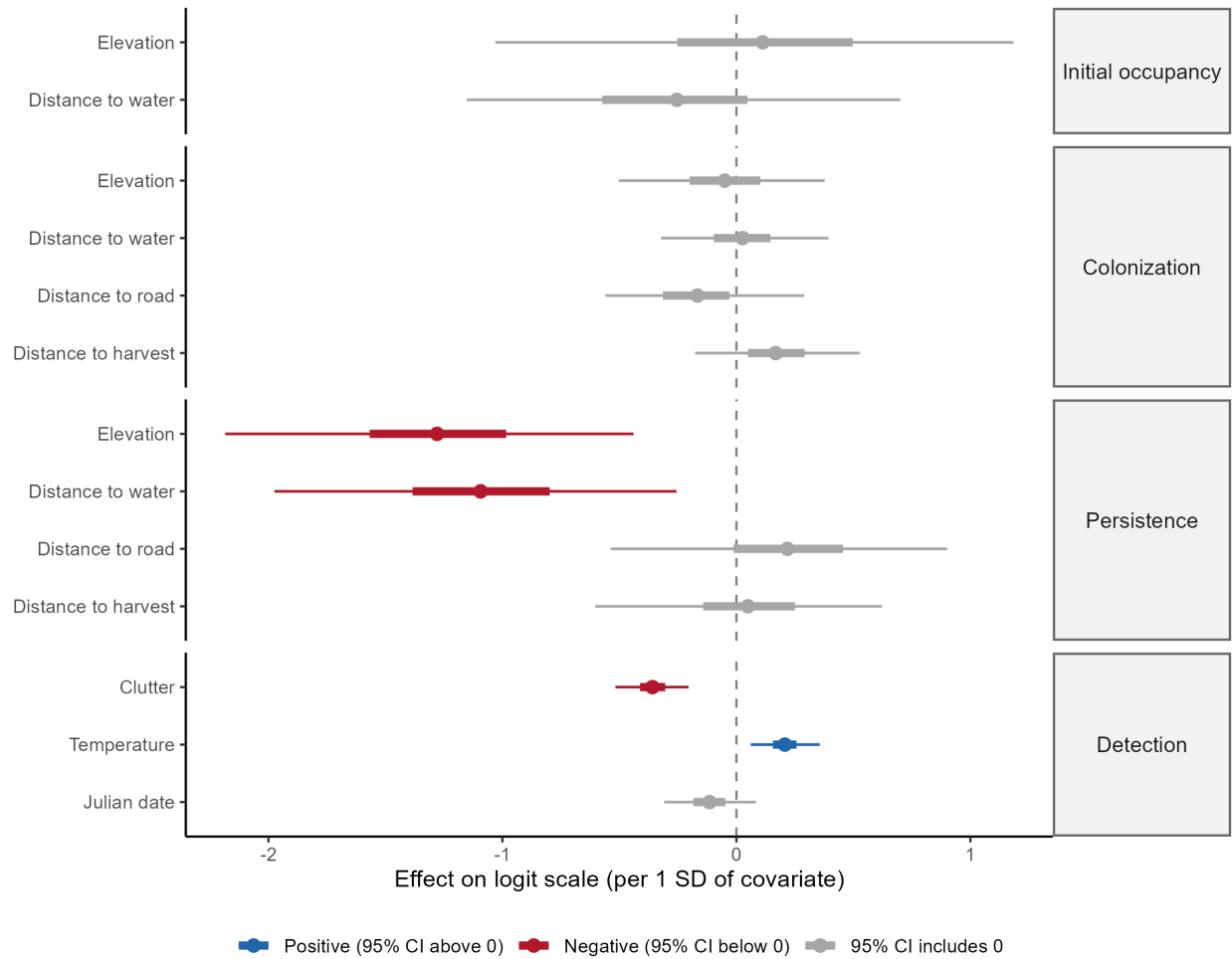
MYCI — Western small-footed myotis: Kootenay

Posterior mean (95% CI) across all 39 sites.



MYCI — Western small-footed myotis: covariate effects

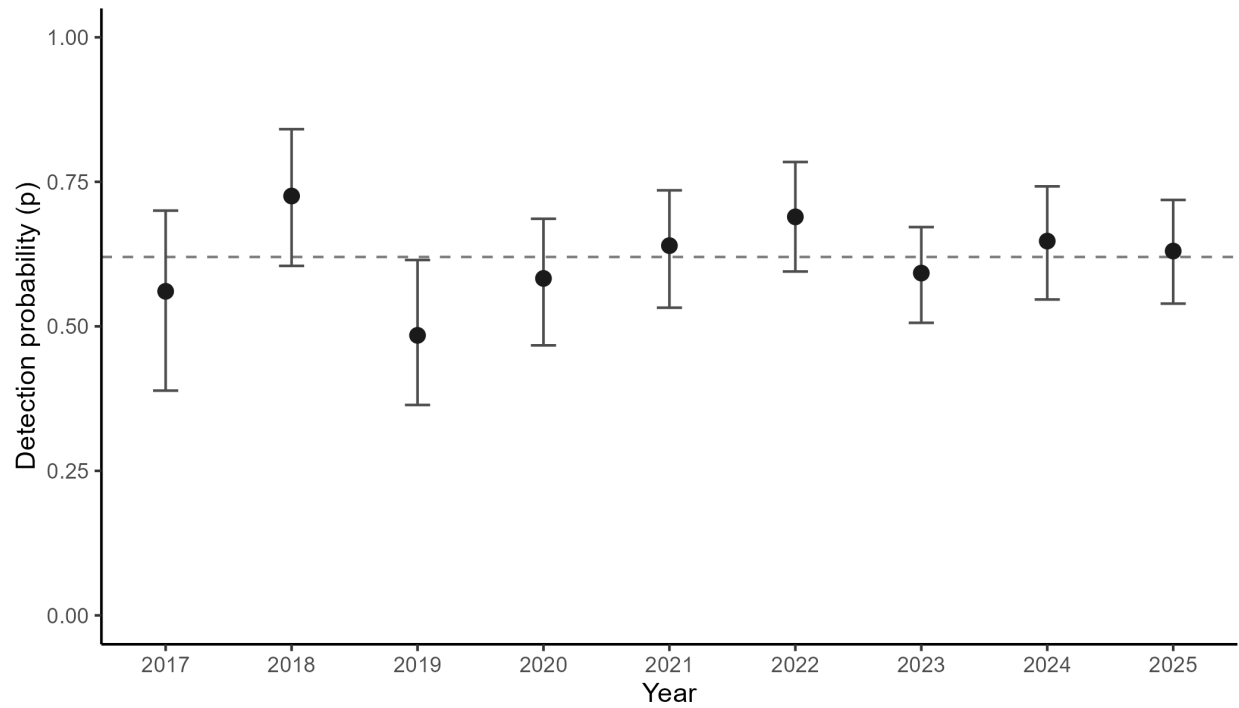
Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.



Distances were $\log(x + 1)$ transformed before standardising.

MYCI — Western small-footed myotis: detection probability by year

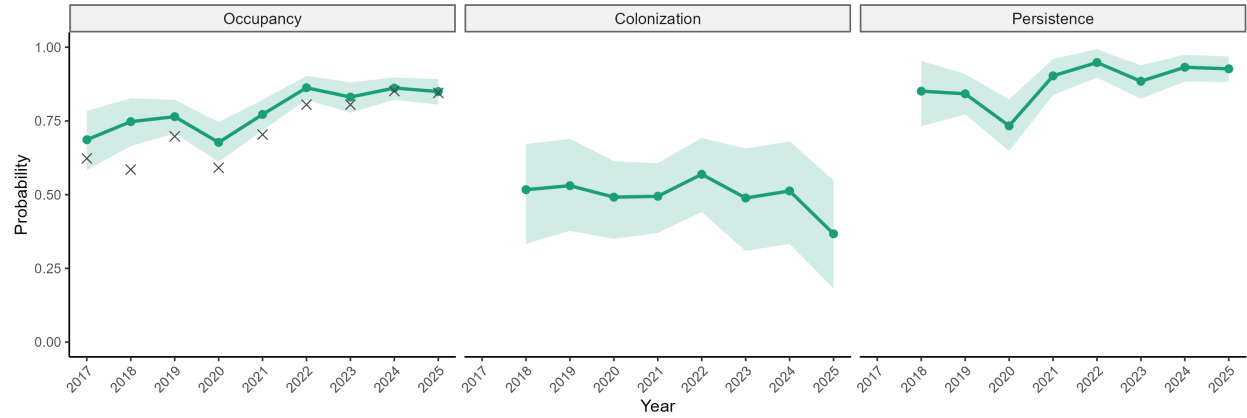
Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.



0.10 Western long-eared Myotis (*Myotis evotis*)

MYEV — Western long-eared myotis: BC-wide dynamics

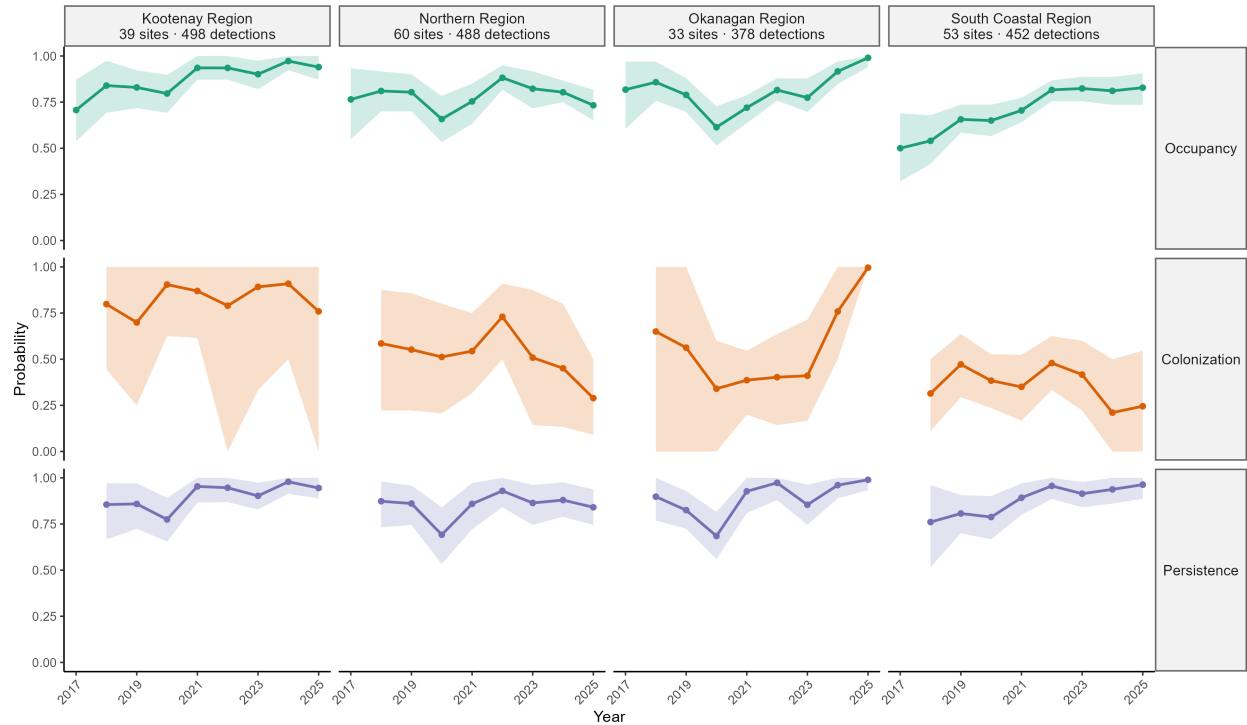
Line = posterior mean (95% CI) across all 185 sites. X = proportion of surveyed sites with a detection.



Colonization and persistence are realised rates from the latent occupancy states: the share of unoccupied sites that became occupied, and of occupied sites that stayed occupied, between consecutive years.

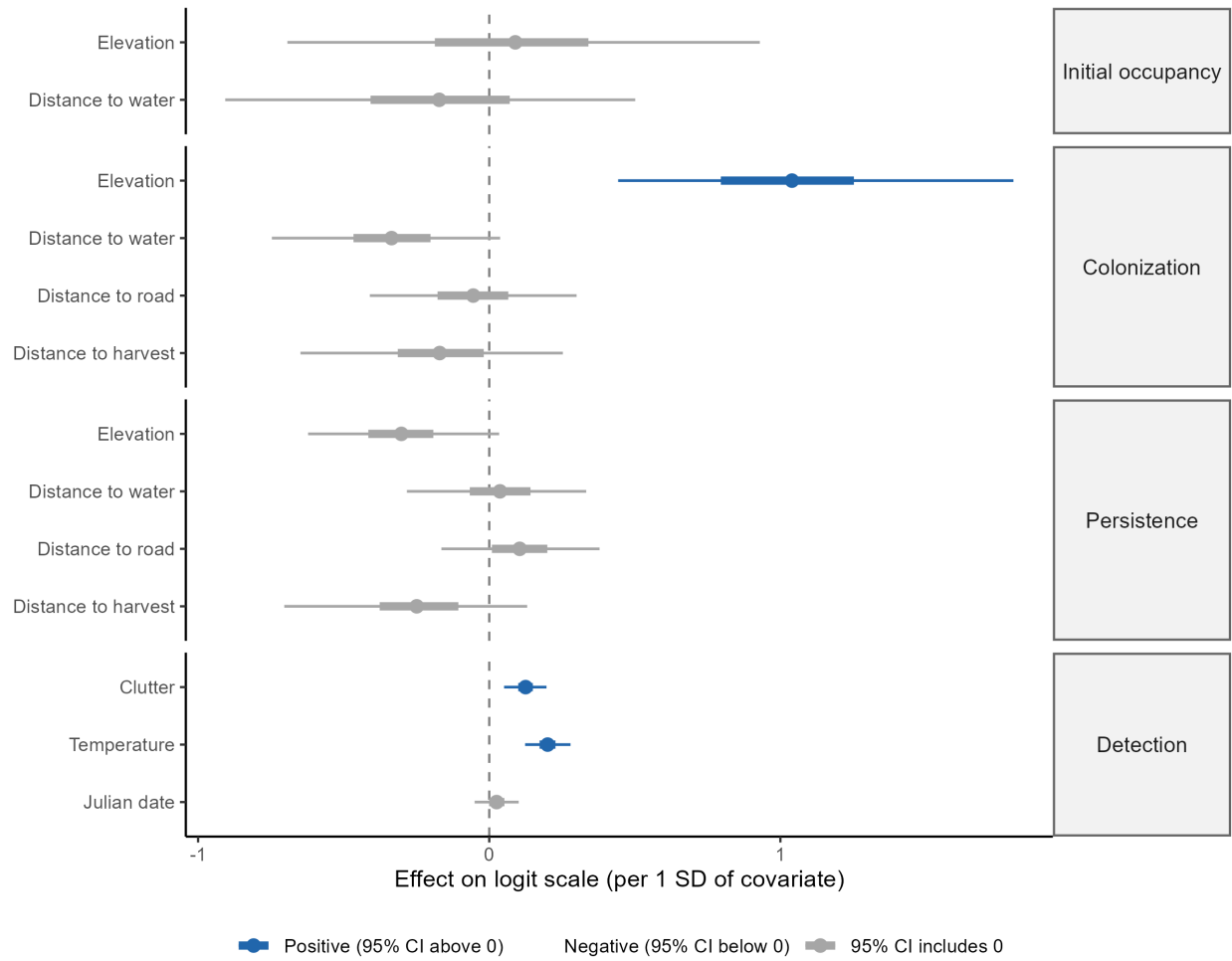
MYEV — Western long-eared myotis: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



MYEV — Western long-eared myotis: covariate effects

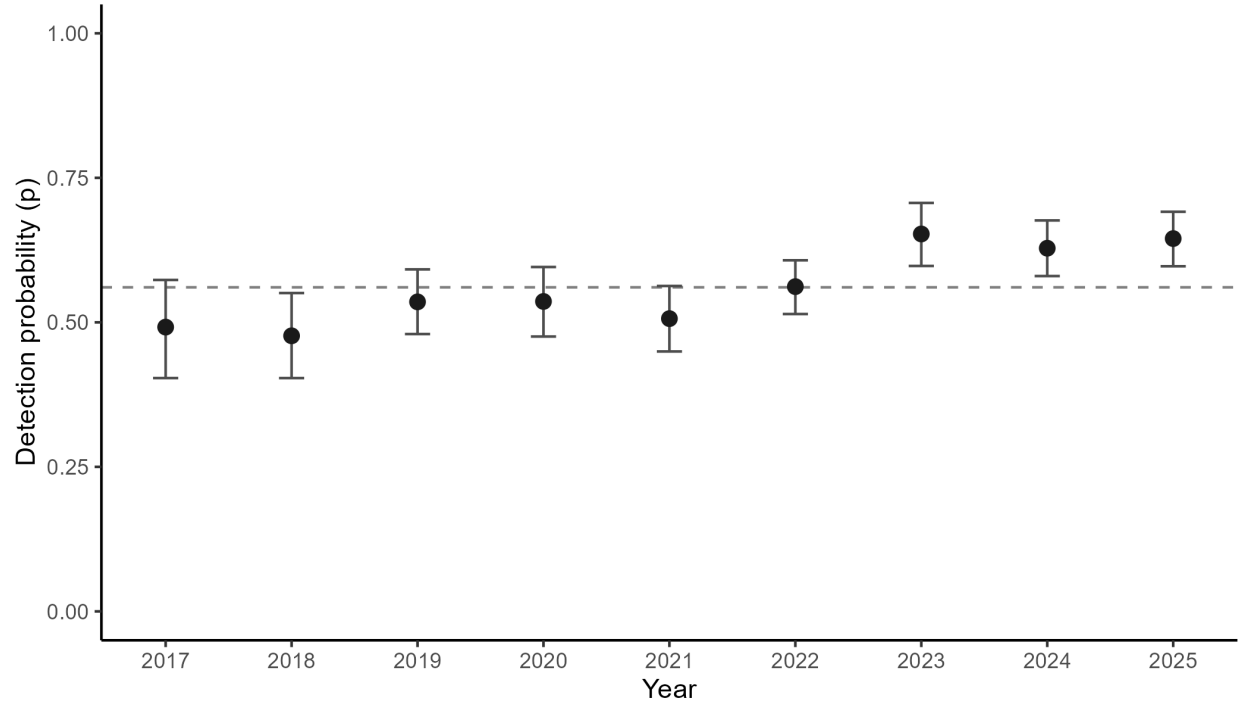
Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.



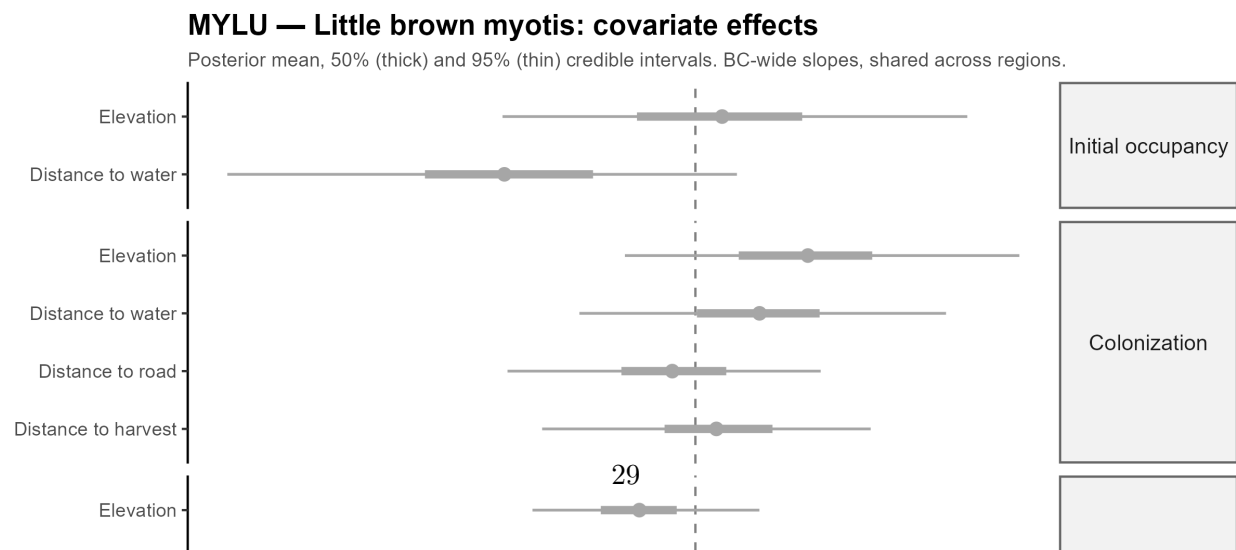
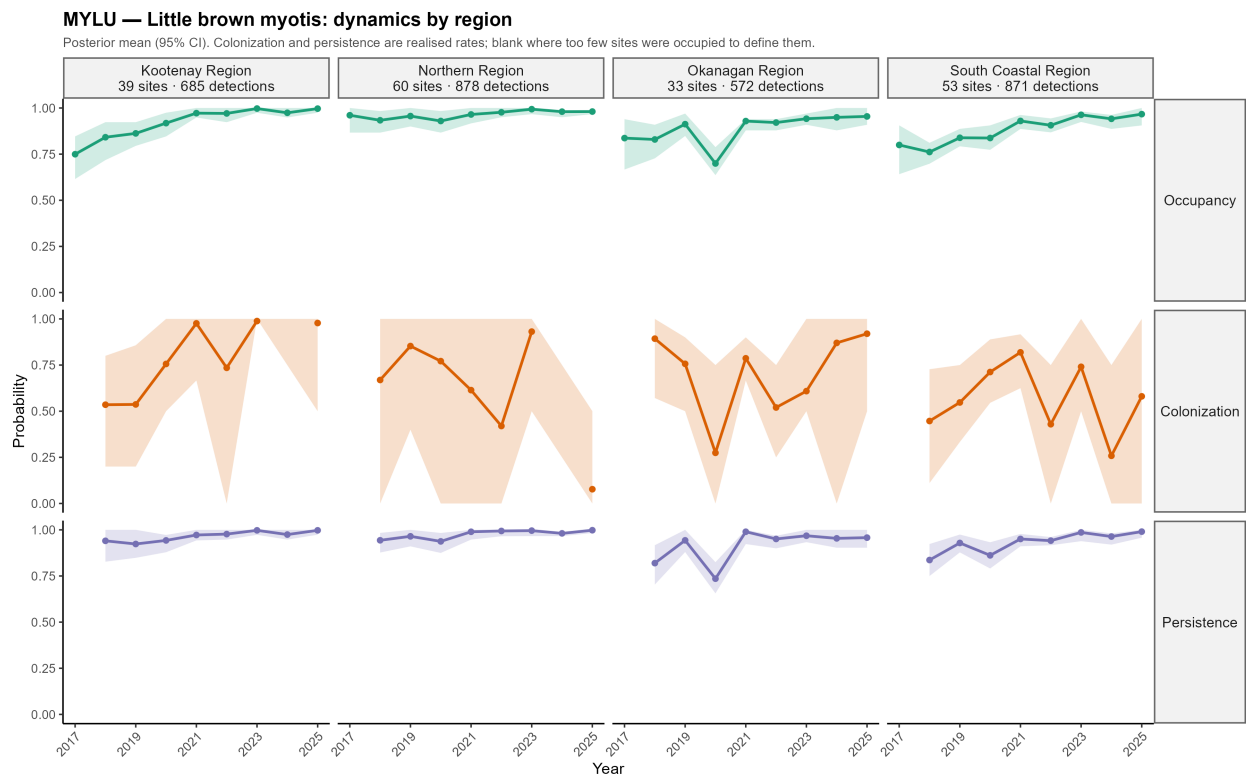
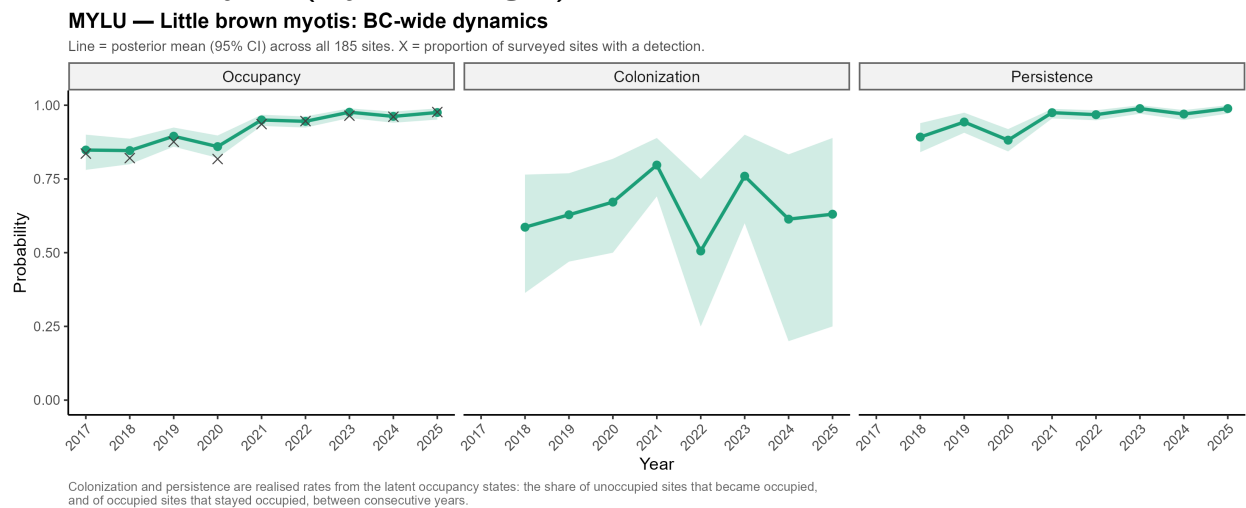
Distances were $\log(x + 1)$ transformed before standardising.

MYEV — Western long-eared myotis: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.

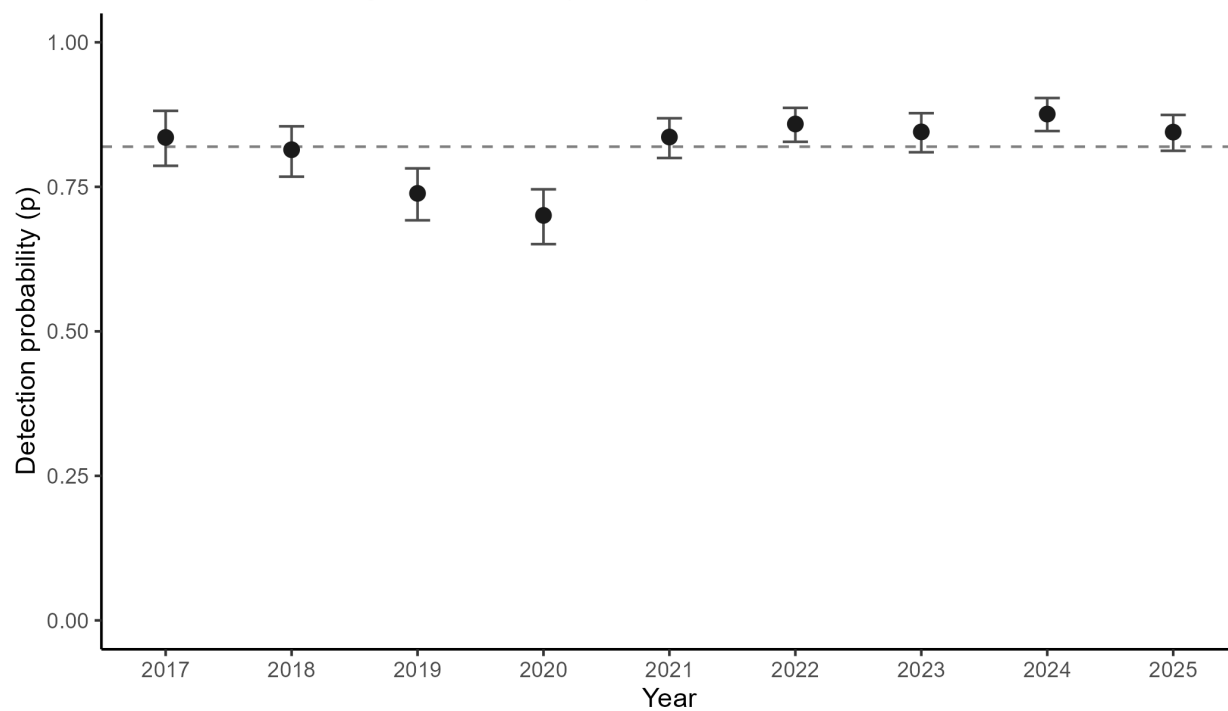


0.11 Little brown Myotis (*Myotis lucifugus*)

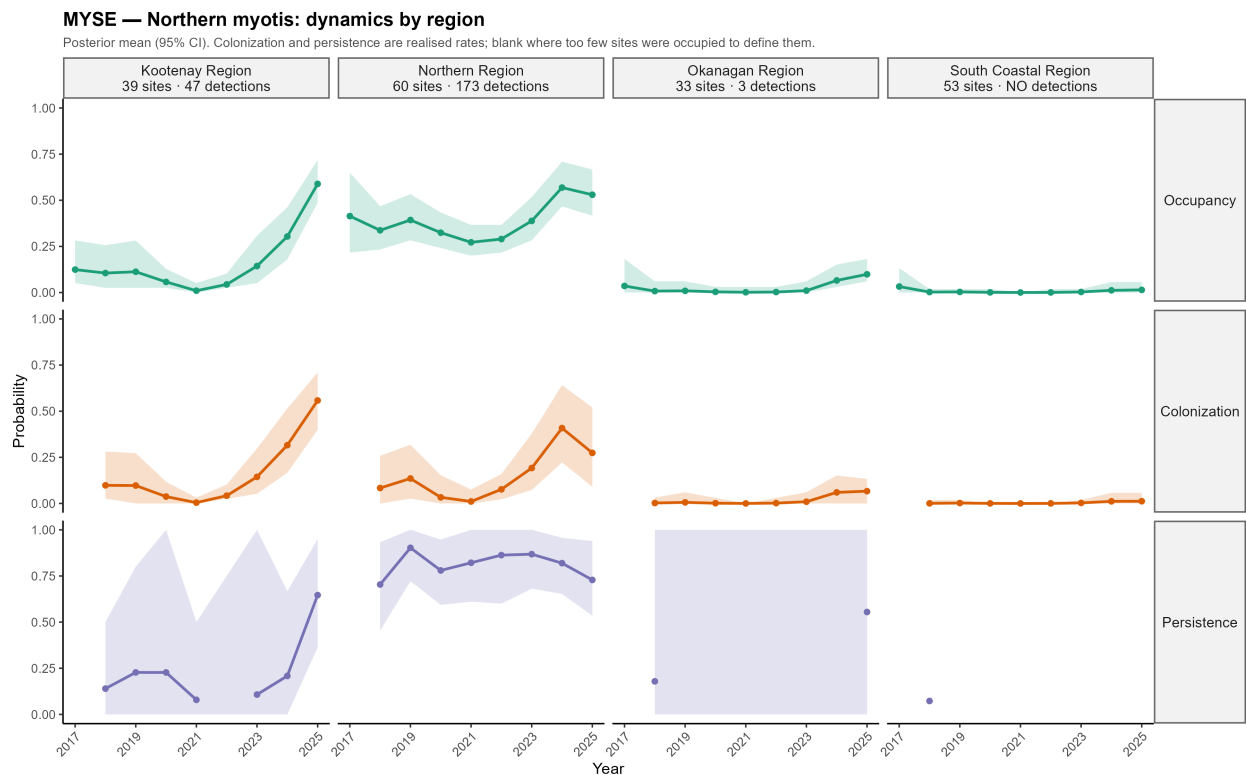
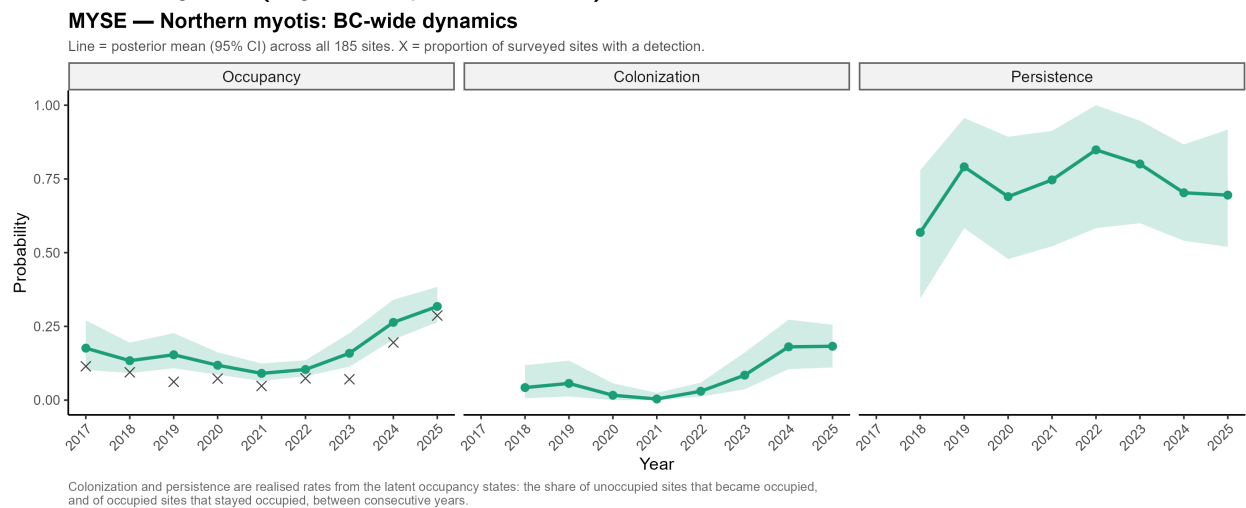


MYLU — Little brown myotis: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.

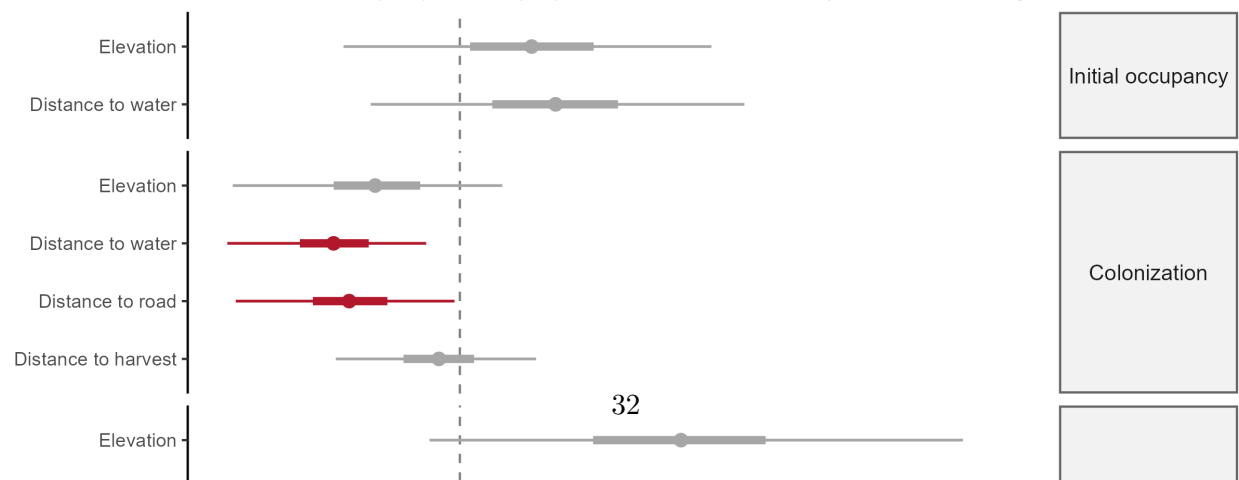


0.12 Northern Myotis (*Myotis septentrionalis*)



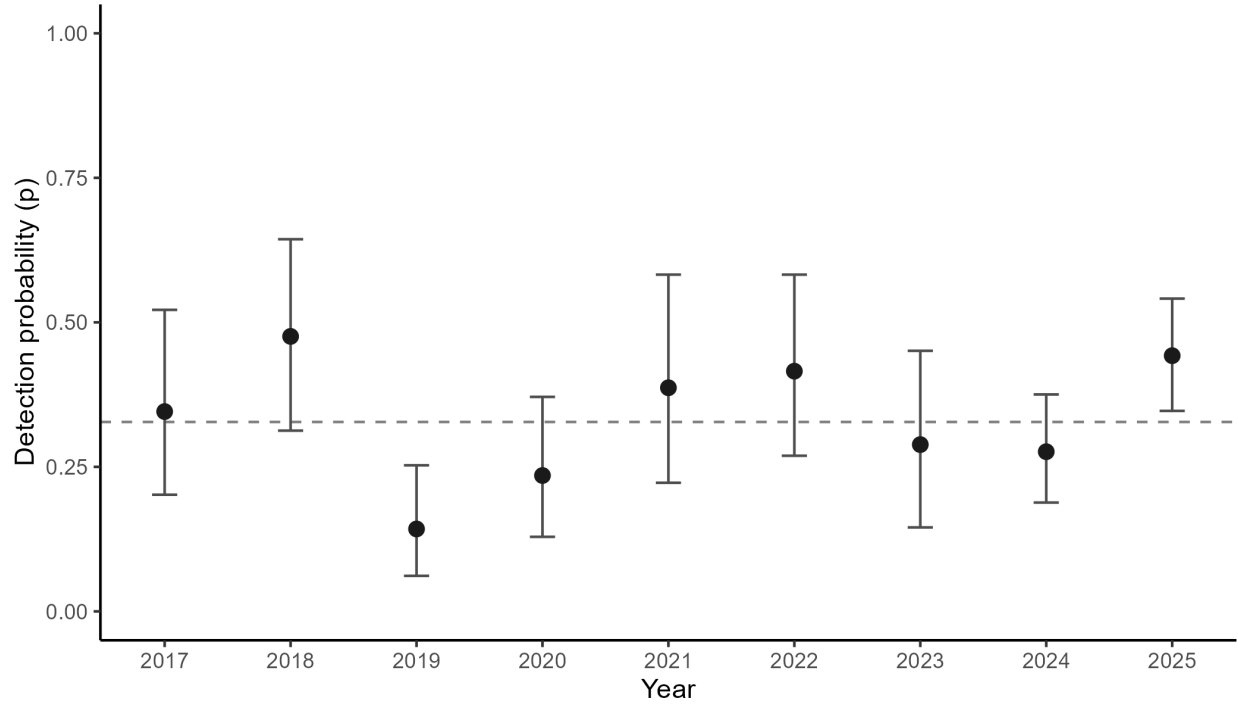
MYSE — Northern myotis: covariate effects

Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.



MYSE — Northern myotis: detection probability by year

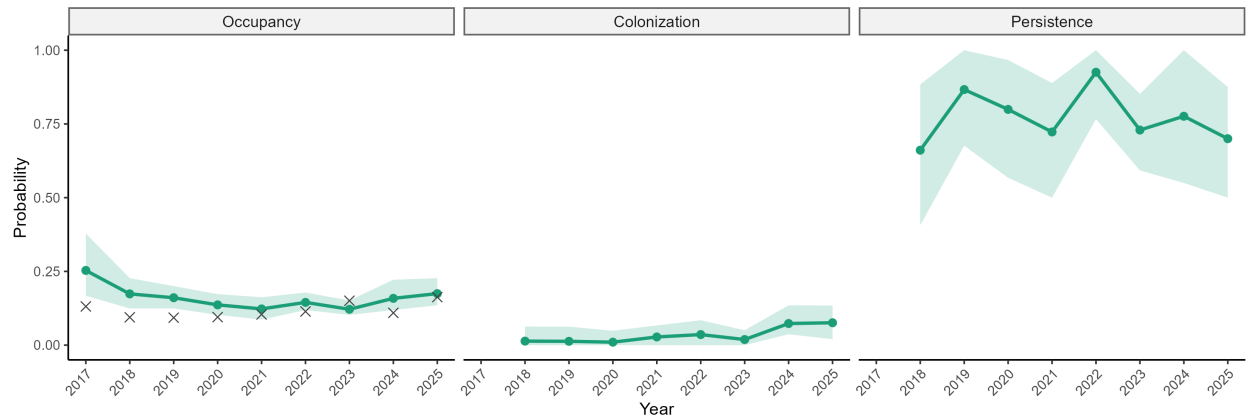
Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.



0.13 Fringed Myotis (*Myotis thysanodes*)

MYTH — Fringed myotis: BC-wide dynamics

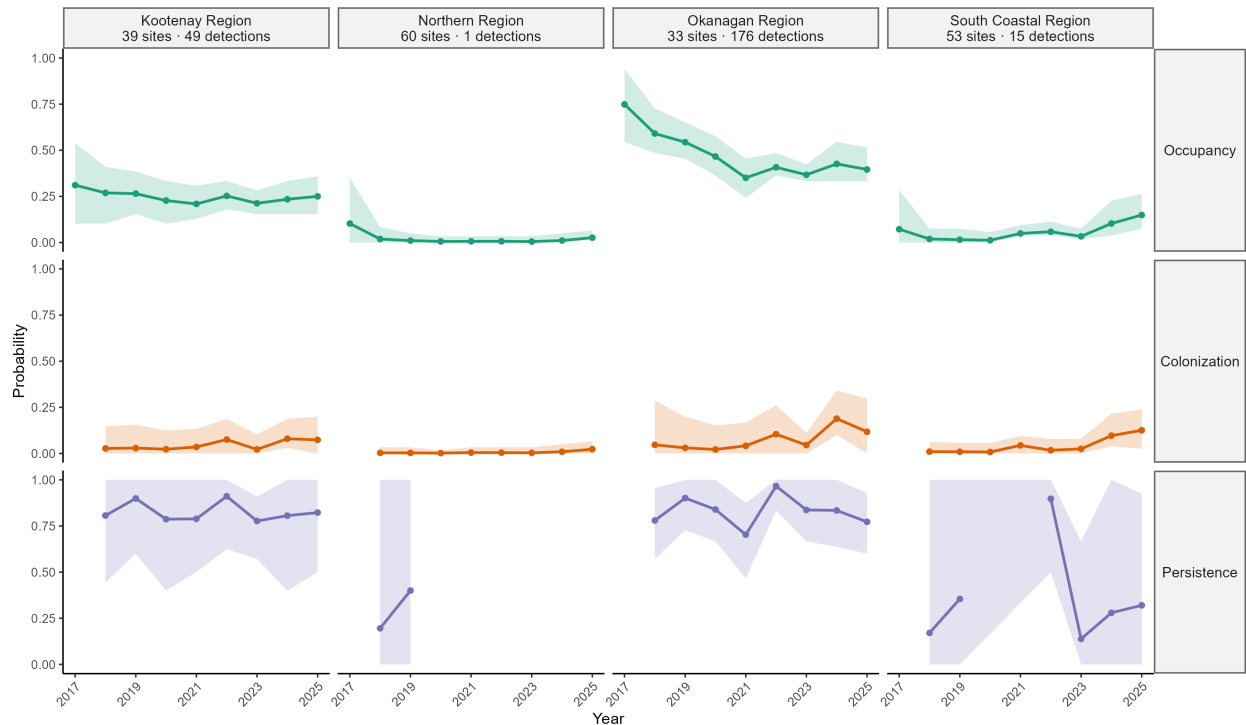
Line = posterior mean (95% CI) across all 185 sites. X = proportion of surveyed sites with a detection.



Colonization and persistence are realised rates from the latent occupancy states: the share of unoccupied sites that became occupied, and of occupied sites that stayed occupied, between consecutive years.

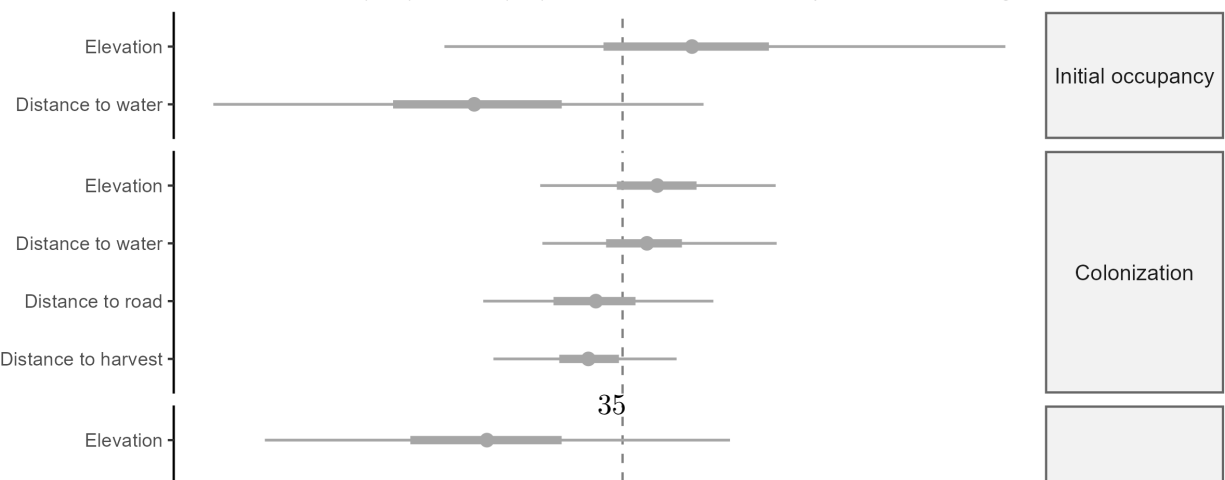
MYTH — Fringed myotis: dynamics by region

Posterior mean (95% CI). Colonization and persistence are realised rates; blank where too few sites were occupied to define them.



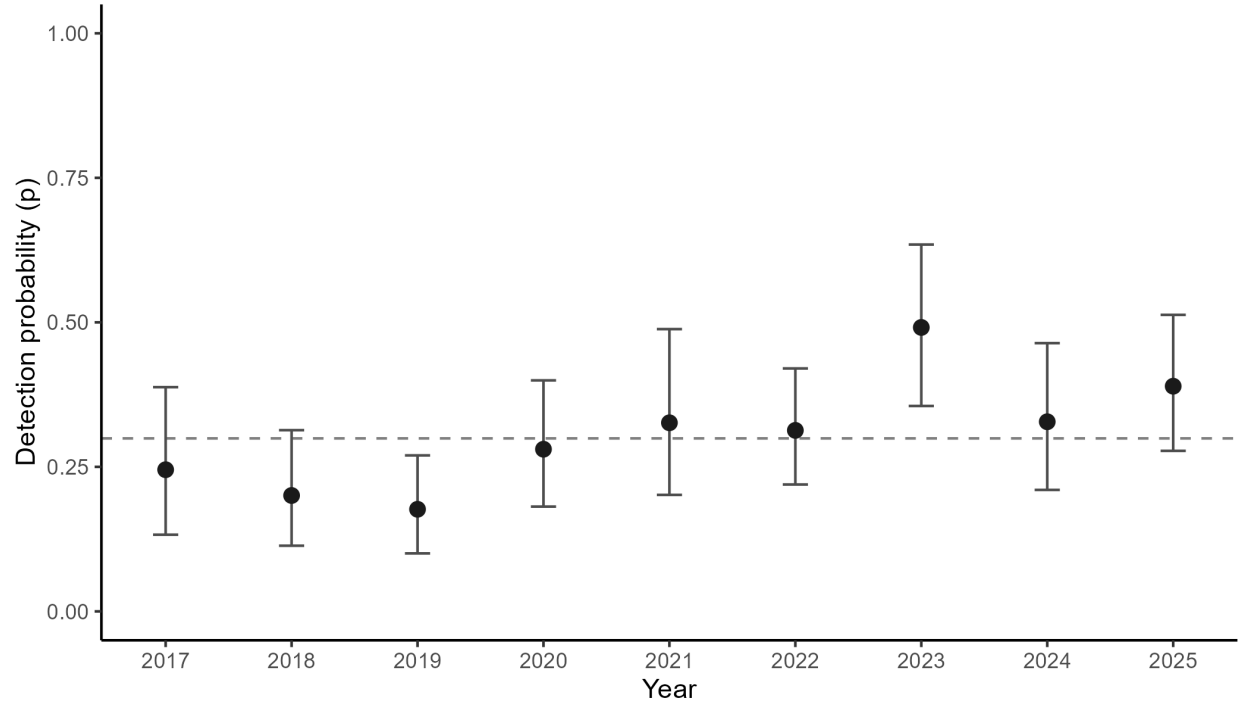
MYTH — Fringed myotis: covariate effects

Posterior mean, 50% (thick) and 95% (thin) credible intervals. BC-wide slopes, shared across regions.

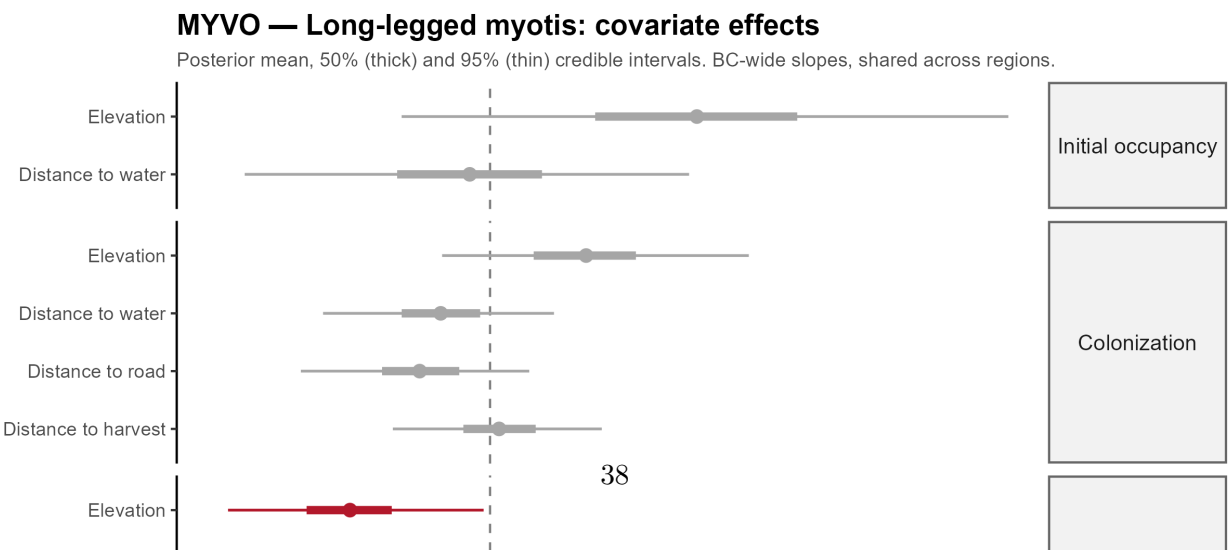
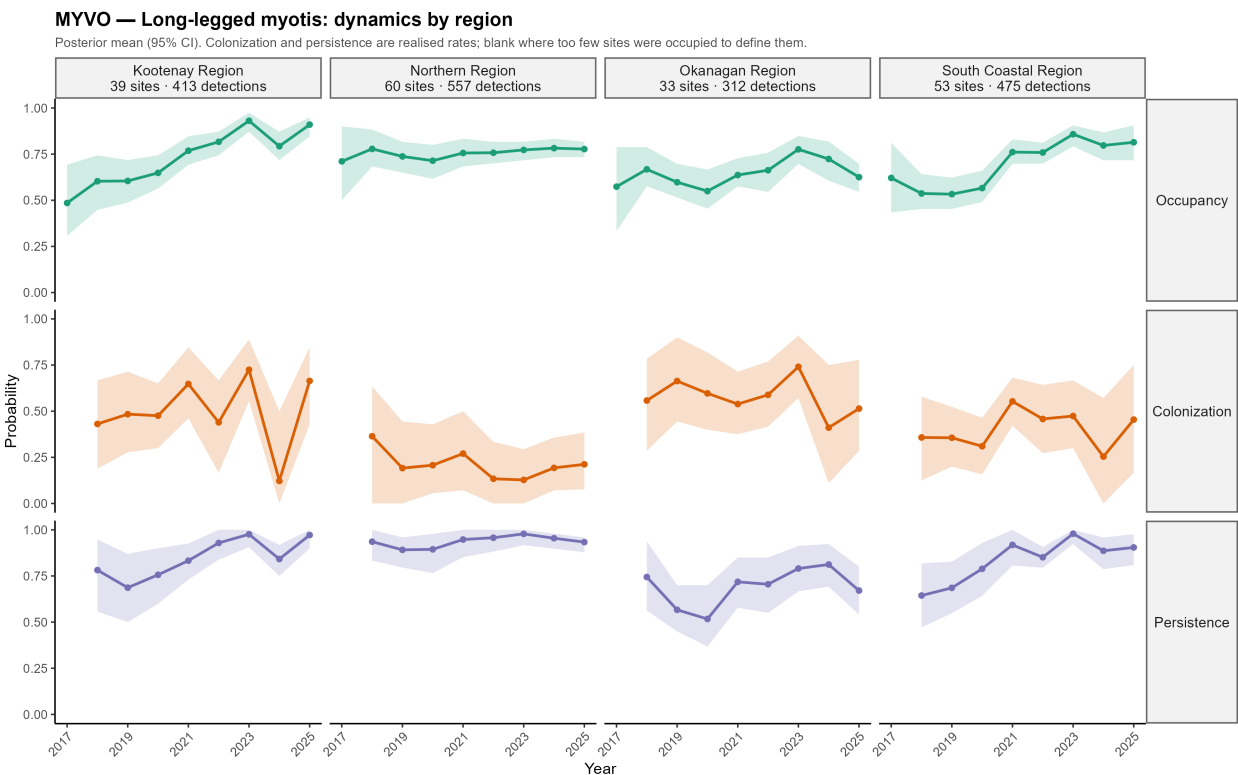
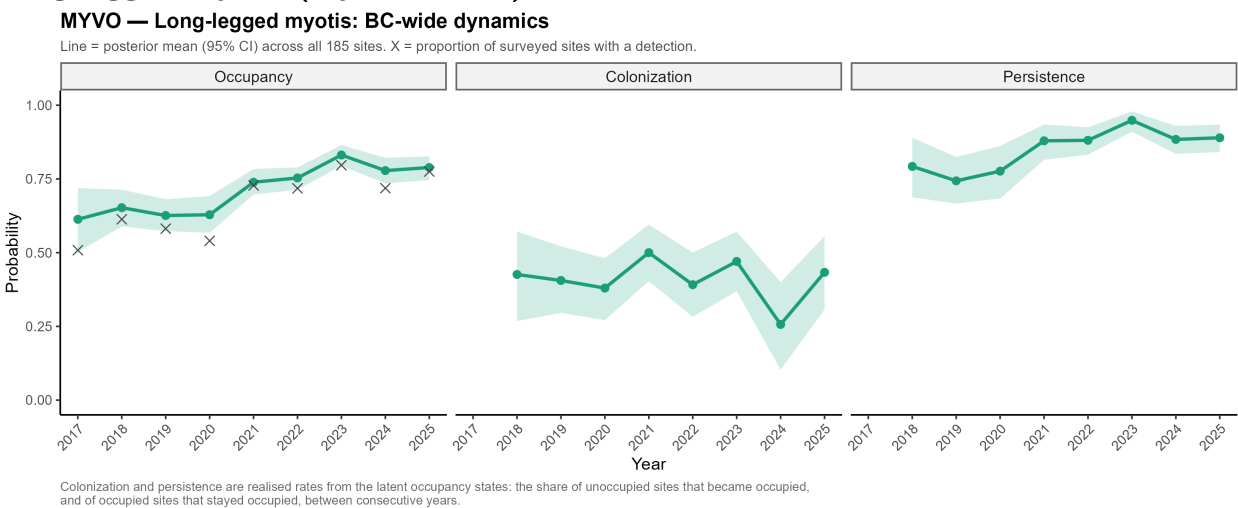


MYTH — Fringed myotis: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.

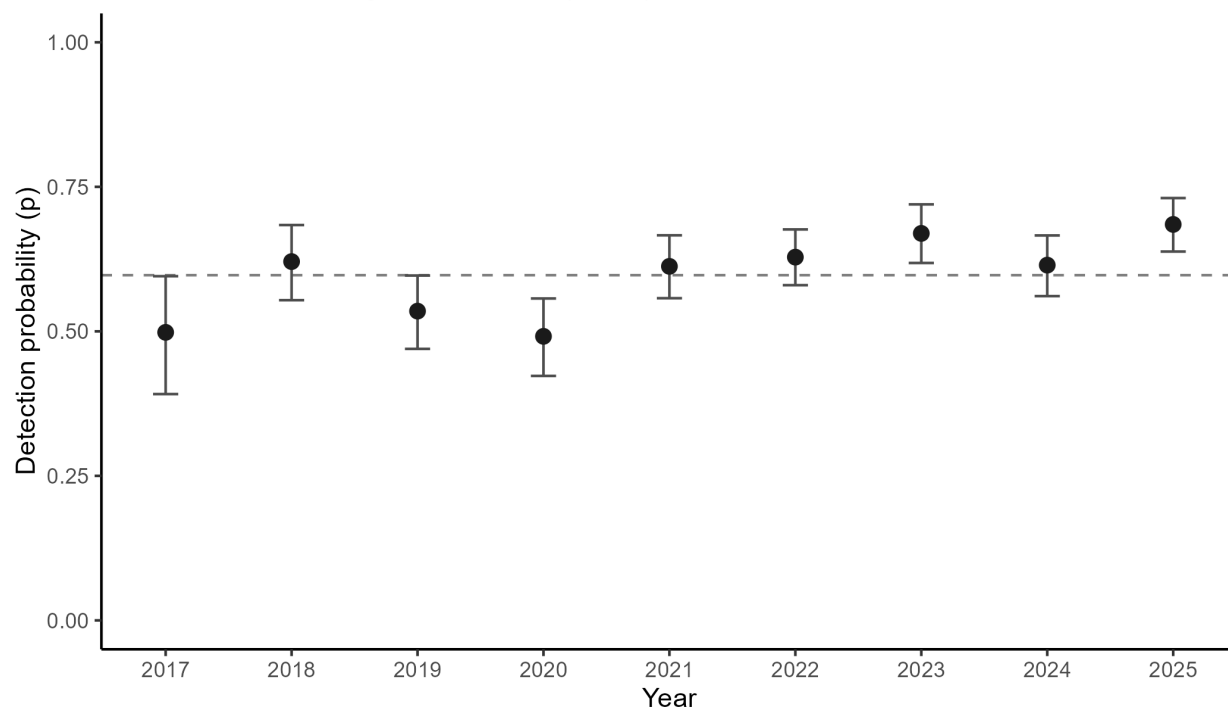


0.14 Long-legged Myotis (*Myotis volans*)

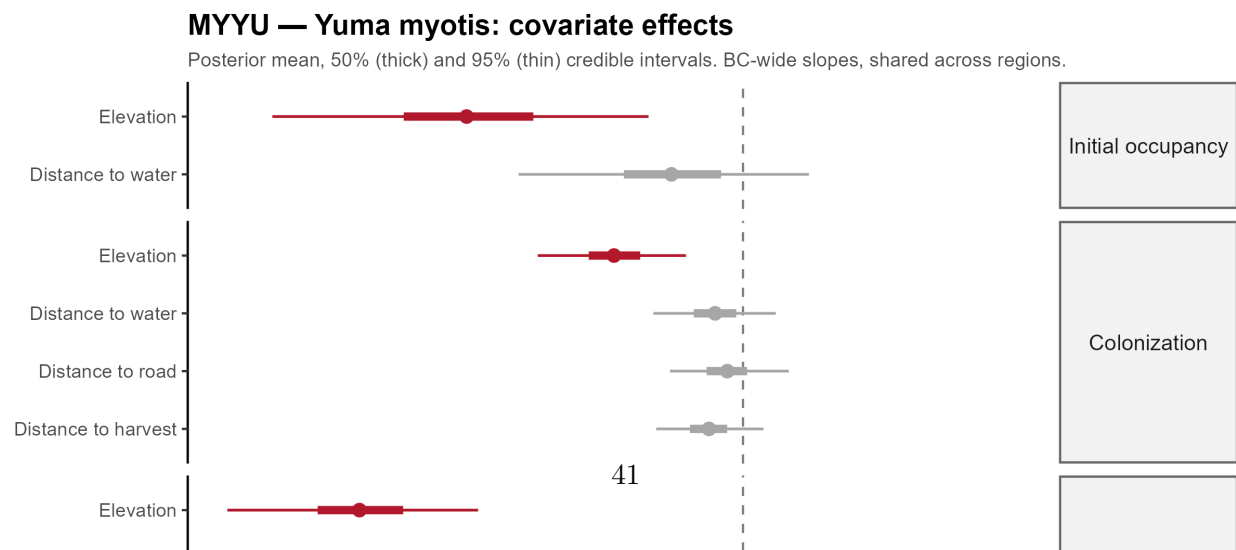
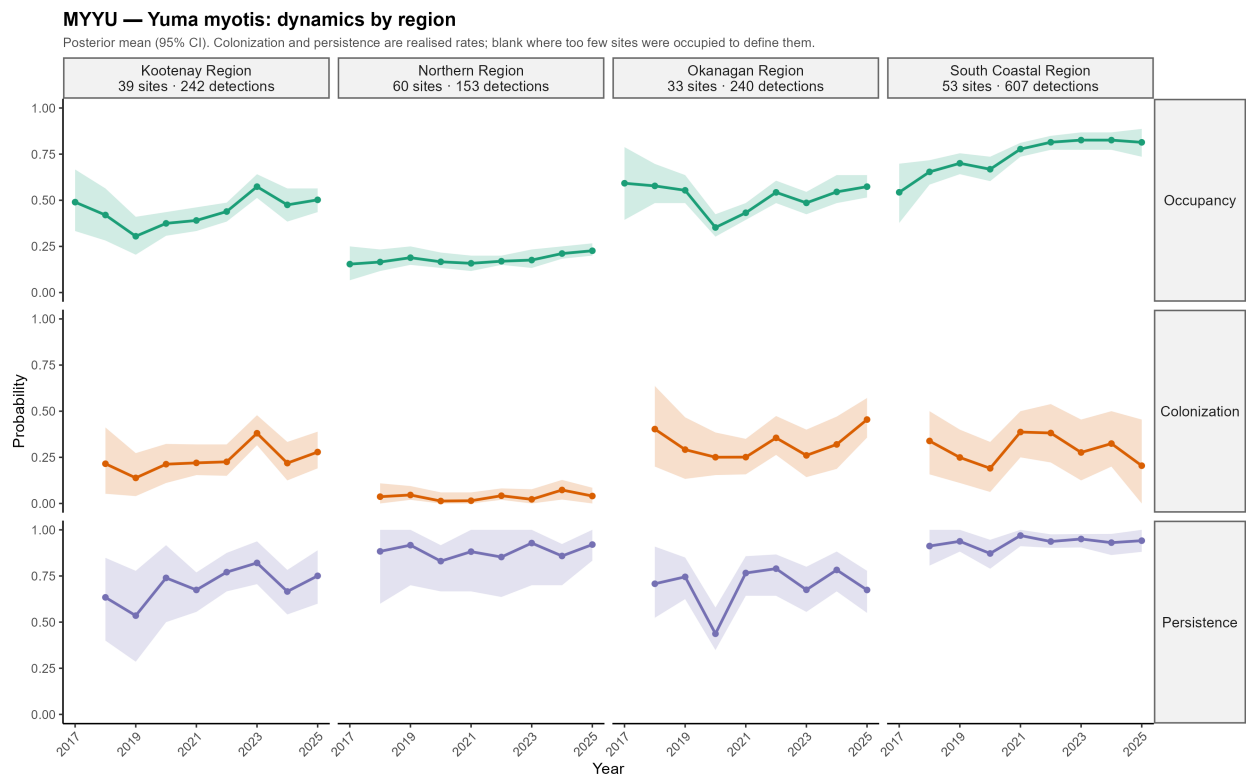
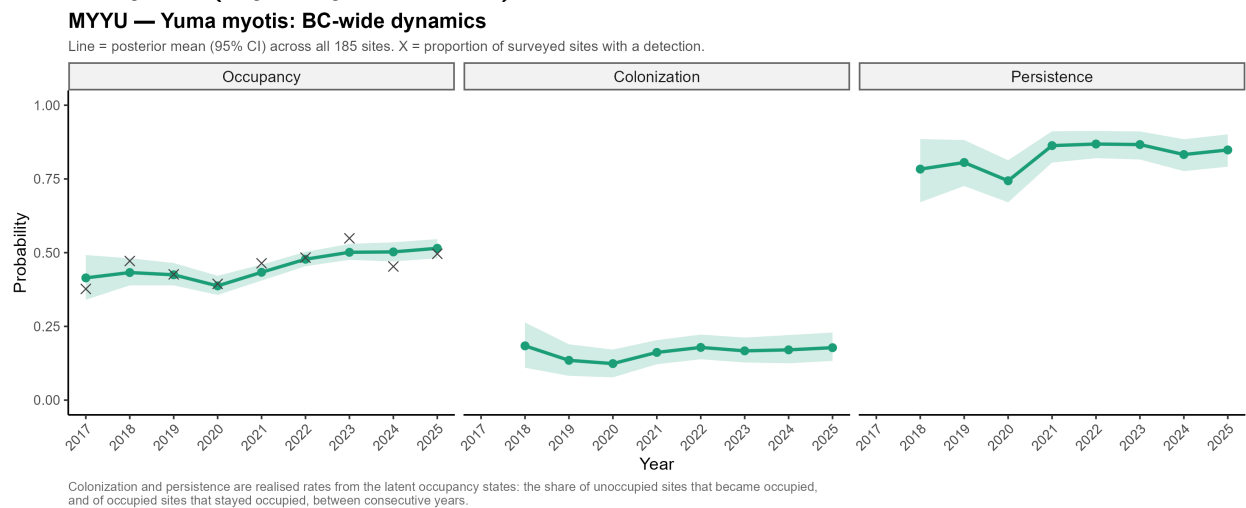


MYVO — Long-legged myotis: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.

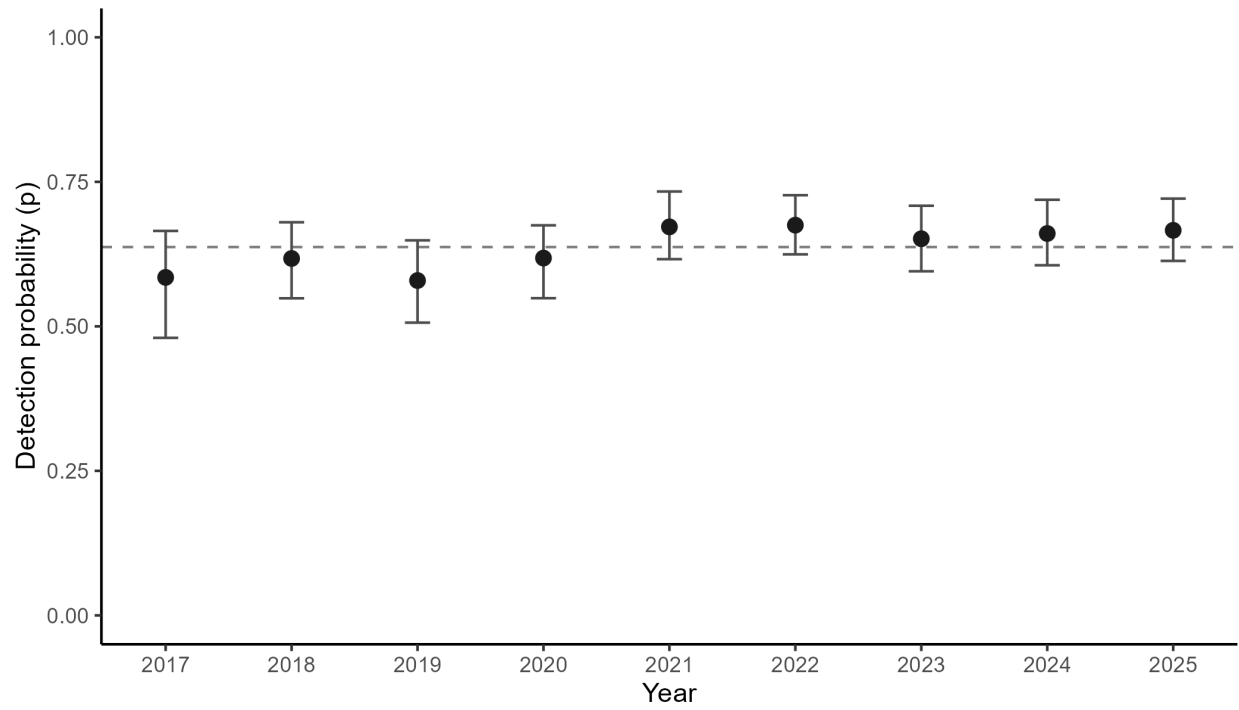


0.15 Yuma Myotis (*Myotis yumanensis*)



MYU — Yuma myotis: detection probability by year

Per-visit detection at average covariate values (95% CI). Dashed line = overall mean.



1 Literature Cited